CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

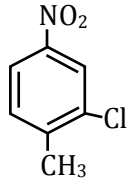
REVISION FLASH CARDS &

MANY MORE <<< CLICK

AIM : FREE EDUCATION TO ALL APUL

Exercise - I Classification and Nomenclature

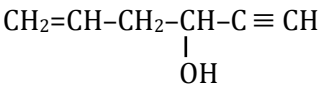
1. IUPAC name of following compound is:



(1) 4-Methyl-5-chloro-nitrobenzene
(2) 2-Chloro-1-methyl-4-nitrobenzene
(3) 1-Chloro-2-methyl-5-nitrobenzene
(4) 1-Methyl-2-chloro-4-nitrobenzene

CCN165

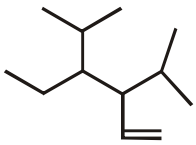
2. IUPAC name for following compound



(1) Hex-1-en-5-yn-4-ol
(2) Hex-5-en-1-yn-3-ol
(3) 4-Hydroxyhex-1-en-5-yne
(4) 3-Hydroxyhex-5-en-1-yne

CCN166

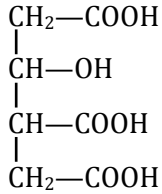
3. IUPAC name for given compound is:-



(1) 3-Ethyl-4-isopropylhex-1-ene
(2) 3, 4-Diisopropylhex-1-ene
(3) 4-Ethyl-3-isopropyl-5-methylhex-1-ene
(4) 3-Ethyl-4-isopropyl-2-methylhex-5-ene

CCN167

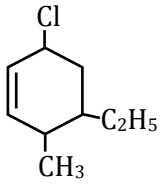
4. IUPAC name for given compound is :-



(1) 3-Hydroxybutane-1,3,5-trioic acid
(2) 3-Hydroxy-4-carboxyhexan-1,6- dioic acid
(3) 2-Hydroxybutane-1,3,4-tricarboxylic acid
(4) 3-Hydroxybutane-1,2,4-tricarboxylic acid

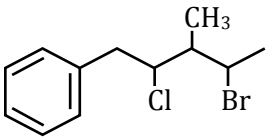
CCN168

5. IUPAC name of:



(1) 3-Chloro-5-ethyl-5-methylcyclohex-1-ene
(2) 2-Chloro-4-ethyl-5-methylcyclohexene
(3) 6-Chloro-4-ethyl-3-methylcyclohex-1-ene
(4) 5-Chloro-3-ethyl-2-methylcyclohex-1-ene

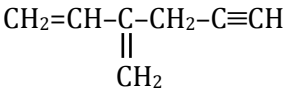
CCN169

6. 

IUPAC name of given compound is :-
(1) 2-Bromo-4-chloro-3-methyl-4-phenylpentane
(2) 2-Chloro-4-Bromo-3-methyl-1-phenylpentane
(3) 4-Bromo-2-chloro-3-methyl-1-phenylpentane
(4) 3-Methyl-2-bromo-4-chloro-5-phenylpentane

CCN170

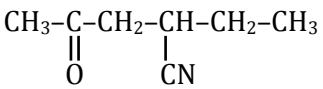
7. IUPAC name of:



(1) 2-Propynylbuta-1, 2-diene
(2) 4-Methylenehex-5-en-1-yne
(3) 3-Methylenehex-1-en-5-yne
(4) 2-Vinylpent-1-en-4-yne

CCN171

8. IUPAC name of the given compound

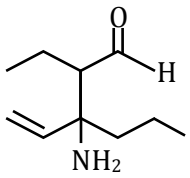


(1) 4-Cyanohex-2-one
(2) 2-Ethyl-4-oxopentanenitrile
(3) 2-Oxo-4-cyanohexane
(4) 2-Oxo-hexane-4-nitrile

CCN172

AIM : FREE EDUCATION TO ALL APUL

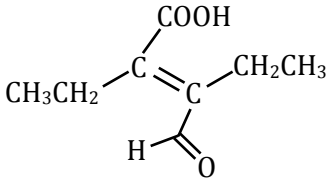
9. The IUPAC name of the compound



(1) 3-Amino-2-ethyl-3-ethenylhexanal
(2) 3-Amino-2-ethyl-3-propylpent-4-en-1-al
(3) 2-Ethyl-3-amino-3-propylpent-4-enal
(4) 4-Propyl-3-oxo-5-hexen-4-amine

CCN173

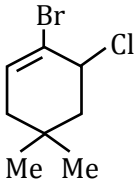
10. Number of carbon in parent carbon chain in the following compound



(1) 4 (2) 5
(3) 6 (4) 3

CCN174

11. The correct IUPAC name of following compound is :



(1) 2-Bromo-3-chloro-5,5-dimethylcyclohex-1-ene
(2) 1-Bromo-6-chloro-4,4-dimethylcyclohex-1-ene
(3) 4-Bromo-5-chloro-1,1-dimethylcyclohex-1-ene
(4) 1-Bromo-2-chloro-4,4-imenthylcyclohex-1-ene

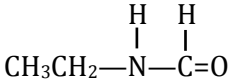
CCN175

12. $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—\underset{\displaystyle CH_3}{\underset{|}{CH}}—$

Radical has IUPAC name–
(1) 4-methylpentyl (2) 1,3-Dimethylbutyl
(3) 2,4-Dimethylbutyl (4) 3-methylpentyl

CCN176

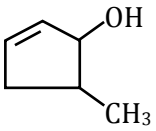
13. One among the following is the correct IUPAC name for the compound



(1) N-Formylaminoethane
(2) N-Ethylformylamine
(3) N-Ethylmethanamide
(4) Ethylaminomethanal

CCN177

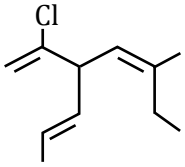
14. The IUPAC name of the compound

 is

(1) 4-Methylcyclopent-1-en-2-ol
(2) 2-Methylcyclopent-4-en-1-ol
(3) 3-Methylcylopent-1-en-2-ol
(4) 5-Methylcyclopent-2-en-1-ol

CCN178

15. IUPAC name of given compound is :-



(1) 2-Chloro-5-methyl-3-(prop-1'-enyl)hepta-1,4-diene
(2) 4-(1'-Chloroethenyl)-6-methylocta-2,5-diene
(3) 5-(1'-Chloroethenyl)-3-methylocta-3,6-diene
(4) 4-(1'-Chloroethenyl)-6-ethylocta-2,5-diene

CCN179

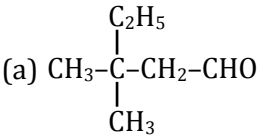
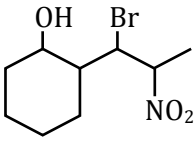
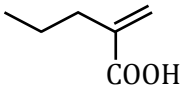
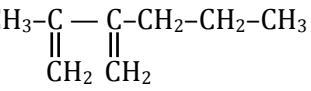
Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**2**	**3**	**4**	**3**	**3**	**3**	**2**	**2**	**2**	**2**	**2**	**3**	**4**	**2**

AIM : FREE EDUCATION TO ALL APUL

Exercise - II Classification and Nomenclature

1. Match list-I with list-II

List-I (Compound)	**List-II** (Number of carbon in PCC)
(a) 	(p) 3C
(b) 	(q) 4C
(c) 	(r) 5C
(d) 	(s) 6C

Choose the correct answer from the options given below :-

(1) a–q b–p c–r d–s (2) a–r b–s c–p d–q

(3) a–p b–q c–r d–s (4) a–r b–p c–s d–q

CCN180

2. Wrong IUPAC name is :-

(1) $CH_3–CH(Cl)–CH(Br)–CH_2–CH(Cl)CH_3$

2,5-Dichloro-3-bromohexane

(2) $CH_3–CH=CH–COOH$

But-2-enoic acid

(3) $(CH_3)_3CCH_2OH$

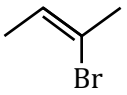
2,2-Dimethylpropan-1-ol

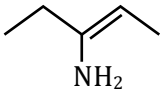
(4) $CH_3CH_2NHCH_3$

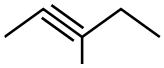
N-Methylethanamine

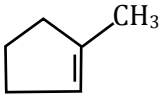
CCN181

3. Correct match is

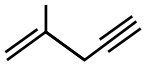
(1)  → 3-Bromo but-2-ene

(2)  → 3-Amino pent-2-ene

(3)  → 2-Methyl but-2-yne

(4)  → 1-Methyl cyclopentene

CCN182

4. The hybridization of C_2, C_3 and C_4  respectively

(1) sp^2, sp^3, sp

(2) sp^3, sp^2, sp

(3) sp^2, sp, sp

(4) sp, sp^3, sp^2

CCN183

5. Which set of IUPAC name is wrong ?

(a) 3-Ethyl-1-pentene

(b) 2-Hydroxy butane

(c) 2-Cyano-3-hydroxybutane

(d) 2-Formyl ethanoic acid

(1) a and c

(2) c and d

(3) a, b, c

(4) b, c and d

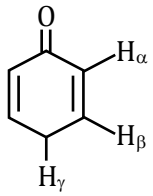
CCN184

Exercise-II ANSWER KEY

Question	1	2	3	4	5
Answer	**2**	**1**	**4**	**1**	**4**

Exercise – I
Isomerism

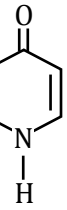
1. Which of the following hydrogen is involved in enolisation of given molecule ?

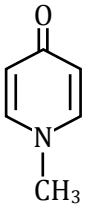


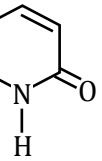
(1) H_α and H_β
(2) H_β
(3) H_γ
(4) Cannot be enolized

CIS118

2. Which among these can exhibit tautomerism?

(a) 

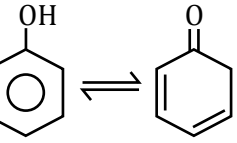
(b) 

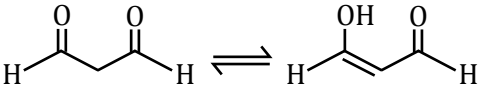
(c) 

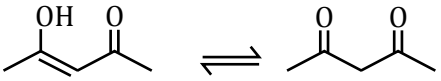
(1) a only (2) b only
(3) c only (4) a and c

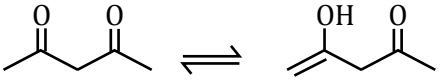
CIS119

3. Which has maximum value of equilibrium constant for given reactions ?

(1) 

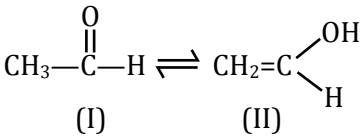
(2) 

(3) 

(4) 

CIS120

4. Which tautomer is more stable :

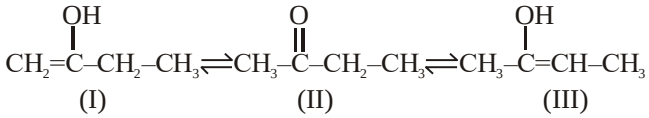


(I) (II)

(1) I (2) II
(3) I = II (4) None of these

CIS121

5. Stability order among these tautomer is:



(I) (II) (III)

(1) I > II > III (2) III > II > I
(3) II > I > III (4) II > III > I

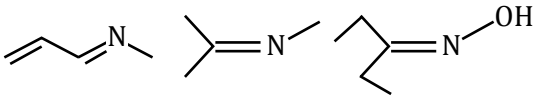
CIS122

6. Which type of strain is present in eclipsed conformation of butane :-

(1) Torsional strain
(2) Steric strain
(3) Both (1) and (2)
(4) None of these

CIS123

7. Which of these will exhibit geometrical isomerism?

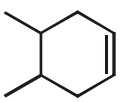


(I) (II) (III)

(1) I (2) II (3) III (4) I and II

CIS124

8. How many geometrical isomers of this compound are possible:

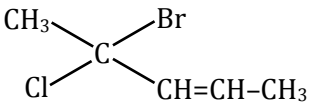


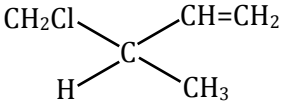
(1) 0 (2) 2 (3) 3 (4) 4

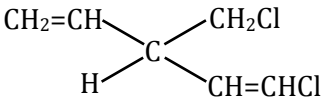
CIS125

AIM : FREE EDUCATION TO ALL APUL

9. Which one among the following will not show geometrical isomerism ?

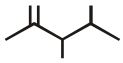
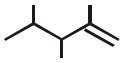
(1) 

(2) 

(3) 

(4) $CH_3CH = CHCH_2CH_3$

CIS126

10.  and  are :-

(1) Positional isomer

(2) Chain isomer

(3) Identical

(4) Metamer

CIS127

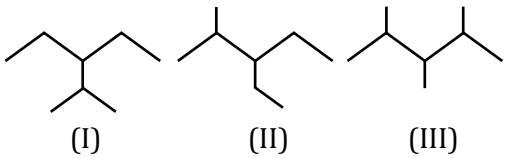
AIM : FREE EDUCATION TO ALL APUL

Exercise-I **ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	10
Answer	**3**	**4**	**2**	**1**	**4**	**3**	**1**	**2**	**2**	**3**

Exercise - II — Isomerism

1. Among these position isomers are



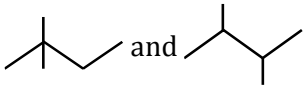
(I) (II) (III)

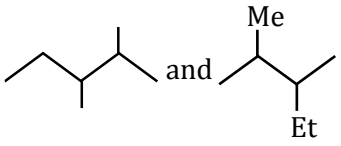
(1) I and II (2) II and III

(3) I and III (4) None

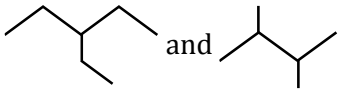
CIS128

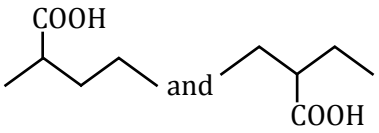
2. Which are chain isomers ?

(1)  and

(2) 

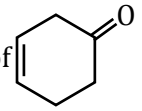
and

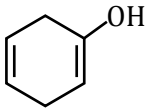
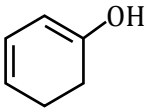
(3)  and

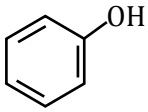
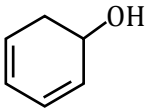
(4) 

and

CIS129

3. Tautomer of  is/are

(a)  (b) 

(c)  (d) 

(1) a and b (2) a, b and c

(3) c and d (4) All

CIS130

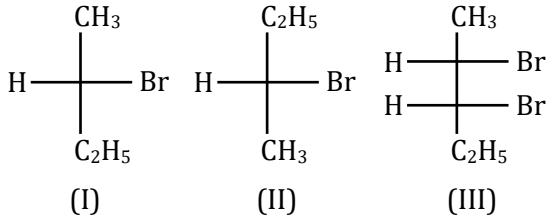
AIM : FREE EDUCATION TO ALL APUL

Exercise-II — ANSWER KEY

Question	1	2	3
Answer	**4**	**4**	**1**

Exercise – I
Optical Isomerism

1. Which of these molecule is optically active?

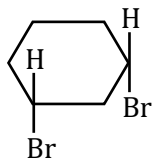


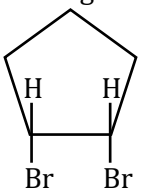
(I) (II) (III)

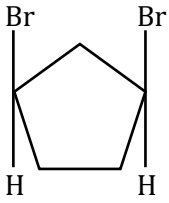
(1) I (2) II
(3) III (4) All of these

COI064

2. Which of the following compounds can have non-superimposable mirror images?

(1) 

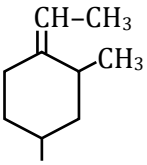
(2) 

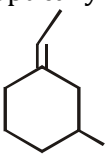
(3) 

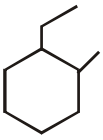
(4) None of the above

COI065

3. Which of the following is optically active ?

(1) 

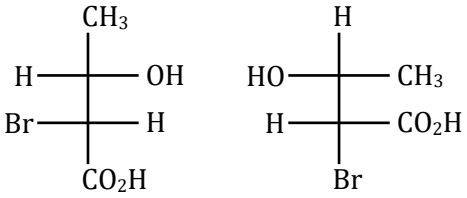
(2) 

(3) 

(4) All of these

COI066

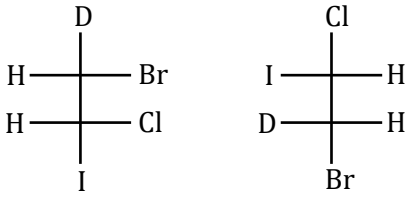
4. The molecules represented by the following two structures are



(1) Identical (2) Enantiomers
(3) Diastereomers (4) Epimers

COI067

5. The two compounds given below are :



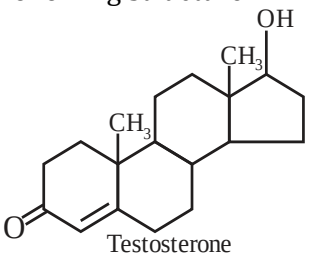
(1) Enantiomer (2) Identical
(3) Meso compound (4) Diastereomers

COI068

6. A racemic mixture is formed by mixing of two:-
(1) Isomeric compounds
(2) Chiral compounds
(3) Enantiomers
(4) Meso compounds

COI069

7. Testosterone, which is a steroidal hormone, has the following structure.



Testosterone

The total number of asymmetric carbon atom/s in testosterone is _
(1) 3 (2) 4 (3) 5 (4) 6

COI070

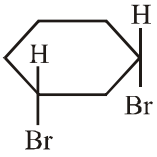
Exercise-I — ANSWER KEY

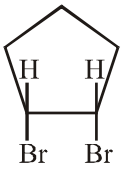
Question	1	2	3	4	5	6	7
Answer	**4**	**4**	**4**	**1**	**1**	**3**	**4**

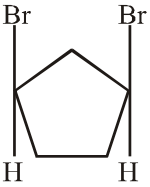
AIM : FREE EDUCATION TO ALL APUL

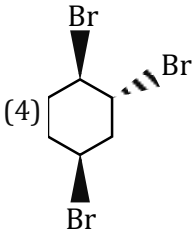
Exercise - II — Optical Isomerism

1. Which of the following compounds can have non-superimposable mirror images?

(1) 

(2) 

(3) 

(4) 

COI071

2. The sum of chiral carbon in

(a) $CH_3-CH=CH-\underset{\underset{Cl}{|}}{CH}-CH=CH-CH_3$

(b) $CH_3-\{CHOH\}_4-CH_3$

(c) $CH_3-\{CHOH\}_3-CH_3$

(1) 8 (2) 4 (3) 7 (4) 6

3. Which of the following compound may not exist as enantiomers ?

(1) $CH_3CH(OH)CO_2H$

(2) $CH_3CH_2CH(CH_3)CH_2OH$

(3) $C_6H_5CH_2CH_3$

(4) $C_6H_5CHClCH_3$

COI072

Exercise-II — ANSWER KEY

Question	1	2	3
Answer	**4**	**1**	**3**

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

Exercise - I

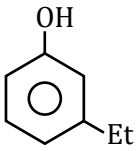
General Organic Chemistry

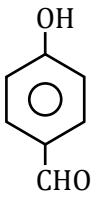
1. K_a of benzoic acid increases if substituent X is bonded at para position, X is :

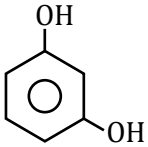
(1) $-OCH_3$ (2) $-CH_3$

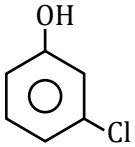
(3) $-CN$ (4) $-NH_2$

CGC236

2. (1) 

(2) 

(3) 

(4) 

Decreasing order of pKa is :

(1) 1 > 2 > 3 > 4 (2) 1 > 3 > 4 > 2

(3) 2 > 4 > 1 > 3 (4) 4 > 3 > 2 > 1

CGC237

3. The correct order of K_a is :

(i) CH_3-CH_2-COOH (ii) $OHC-CH_2-COOH$

(iii) $Ph-CH_2-COOH$

(1) iii > ii > i (2) i > ii > iii

(3) ii > iii > i (4) i > iii > ii

CGC238

4. Decreasing order of stability of the following carbocation ?

(A) $m-(HO)-C_6H_4-\overset{\oplus}{C}H_2$

(B) $p-(HO)-C_6H_4-\overset{\oplus}{C}H_2$

(C) $p-(CHO)-C_6H_4-\overset{\oplus}{C}H_2$

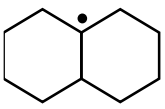
(D) $C_6H_5-\overset{\oplus}{C}H_2$

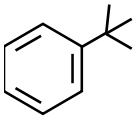
(1) B > A > D > C (2) B > D > A > C

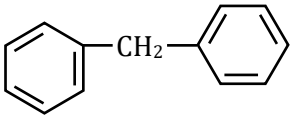
(3) B > D > C > A (4) D > C > B > A

CGC239

5. Which of the following compound will show Hyperconjugation?

(1) 

(2) 

(3) 

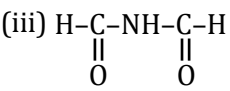
(1) 2 only (2) 1 and 3 only

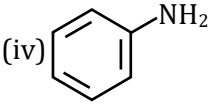
(3) 1 and 2 (4) All of these

CGC240

6. Correct order of pK_b is

(i) $CH_3-CH_2-NH_2$ (ii) $Ph-CH_2-NH_2$

(iii) 

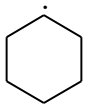
(iv) 

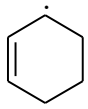
(1) i > ii > iii > iv (2) iii > iv > i > ii

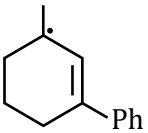
(3) iii > iv > ii > i (4) ii > i > iii > iv

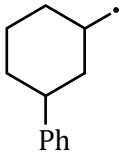
CGC241

7. Which of the following carbon free radical is most stable ?

(1) 

(2) 

(3) 

(4) 

CGC242

8. Correct order of basic strength in gas phase is

(I) CH_3-NH_2 (II) $(CH_3)_2NH$

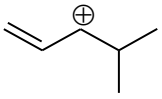
(III) $(CH_3)_3N$ (IV) NH_3

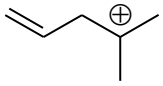
(1) I > II > III > IV (2) II > I > III > IV

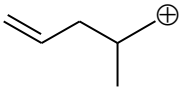
(3) III > II > I > IV (4) II > III > I > V

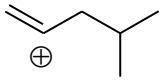
CGC243

9. Which carbocation is most stable ?

(1) 

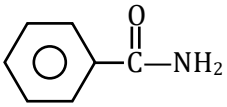
(2) 

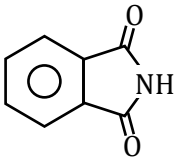
(3) 

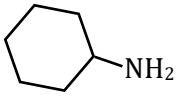
(4) 

CGC244

10. Arrange the following in order of their decreasing basic strength

(I) 

(II) 

(III) 

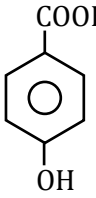
(1) II > III > I (2) I > II > III

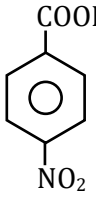
(3) III > I > II (4) I > III > II

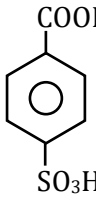
CGC245

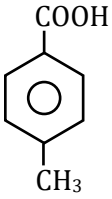
AIM : FREE EDUCATION TO ALL APUL

11. Which of the following is strongest acid ?

(1) 

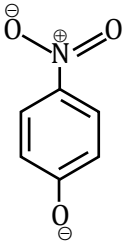
(2) 

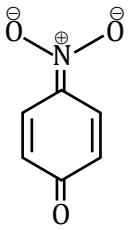
(3) 

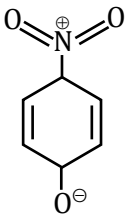
(4) 

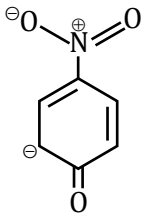
CGC246

12. The most unlikely representation of resonance structure of p-nitrophenoxide is

(1) 

(2) 

(3) 

(4) 

CGC247

13. Which of the following carbanion is most stable?

(1) $H_3C-\overset{\ominus}{C}H_2$ (2) $H_2C=CH-\overset{\ominus}{C}H_2$

(3) $H_3C-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{C}H_2$ (4) $H_3C-\overset{NH}{\overset{\|}{C}}-\overset{\ominus}{C}H_2$

CGC248

14. Which of the following does not represent a correct set of nucleophile–

(1) $HS^{\ominus}$, $C_2H_5O^{\ominus}$

(2) $(CH_3)_3N$, $\overset{\ominus}{\ddot{N}}H_2$, $\overset{\ominus}{C}N$,

(3) SO_3, H_2O, $\overset{\ominus}{\ddot{C}}H_3$

(4) CH_3-S-CH_3, $\overset{\ominus}{\ddot{N}}O_2$, $\overset{\ominus}{I}$

CGC249

15. Which of the following is correct regarding formation of reactive intermediate in given heterolytic bond cleavage:

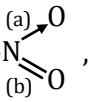
(1) $CH_3-S-CH_3 \longrightarrow \overset{\ominus}{\ddot{C}}H_3 + CH_3-\overset{\oplus}{S}$

(2) $CH_3-Cu \longrightarrow \overset{\ominus}{\ddot{C}}H_3 + \overset{\oplus}{C}u$

(3) $CH_3-Cl \longrightarrow \overset{\ominus}{\ddot{C}}H_3 + \overset{\oplus}{C}l$

(4) (1) & (2) both

CGC250

16. In nitromethane  ,

two N – O bond lengths are:

(1) a = b (2) a > b

(3) b > a (4) Cannot be compared

CGC251

17. Which of the following shows +R effect:

(1)  (2) – Cl

(3) – COOH (4) – CONHR

CGC252

18. Which of the following compounds does not show electromeric effect ?

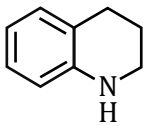
(1) CH_3-CN (2) CH_3-CHO

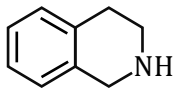
(3) $CH_2=CH_2$ (4) CH_3-Cl

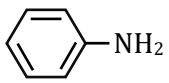
CGC253

19. Which of the following is most basic ?

(1) 

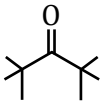
(2) 

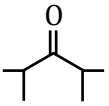
(3) 

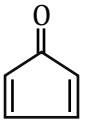
(4) 

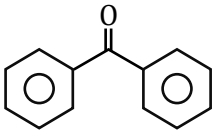
CGC254

20. Tautomerism is shown by :-

(1) 

(2) 

(3) 

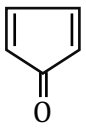
(4) 

CGC255

AIM : FREE EDUCATION TO ALL APUL

21. Which is not anti aromatic ?

(1) 

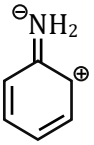
(2) 

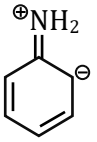
(3) 

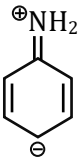
(4) 

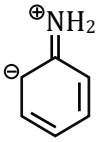
CGC256

22. Which is not the resonance structure of aniline?

(1) 

(2) 

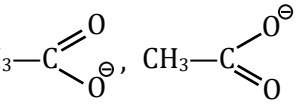
(3) 

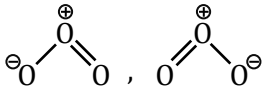
(4) 

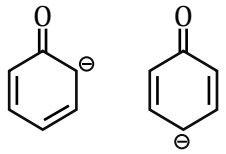
CGC257

23. Pairs of related chemical species are given below. Which pair is not resonating structures?

(1) $CH_3COCH_2COOC_2H_5$, $CH_3C(OH)=CHCOOC_2H_5$

(2) 

(3) 

(4) 

CGC258

24. Which one of the following has all the effects, namely inductive, mesomeric and hyperconjugative?

(1) CH_3Cl

(2) $CH_3CH=CH_2$

(3) $CH_3CH=CHCCH_3$ (O)

(4) $CH_2=CH-CH=CH_2$

CGC259

25. Which among the following alkenes will be most stable:

(1) Ethene

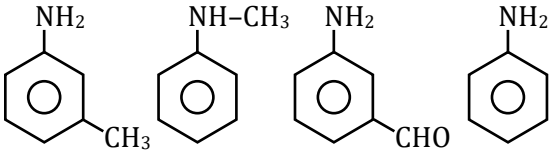
(2) 2-Methylpropene

(3) 2,3-Dimethyl-2-butene

(4) 2-butene

CGC260

26. Correct order of pK_b is ?



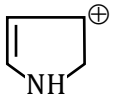
(I) (II) (III) (IV)

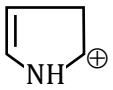
(1) II < I < IV < III (2) I < II < IV < III

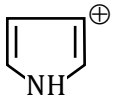
(3) III < IV < I < II (4) III < IV < II < I

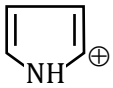
CGC261

27. Which cation is maximum stable ?

(1) 

(2) 

(3) 

(4) 

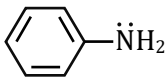
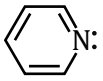
CGC262

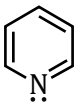
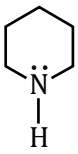
Exercise-I ANSWER KEY

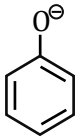
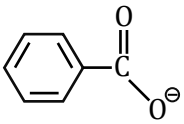
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**3**	**2**	**2**	**3**	**3**	**3**	**1**	**3**	**3**	**3**	**3**	**3**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	**1**	**2**	**4**	**3**	**2**	**1**	**1**	**1**	**3**	**3**	**1**	**1**			

Exercise - II General Organic Chemistry

1. Select the correct statement :-

(1)  is more basic than 

(2)  is more basic than 

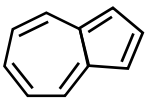
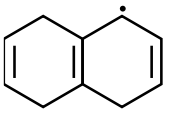
(3)  is more basic than 

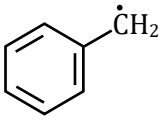
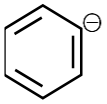
(4) All of them

CGC263

2. Which is/are aromatic

(a)  (b) 

(c)  (d) 

(e)  (f) 

Correct set is

(1) a, b, c and e (2) b, c and e

(3) b, e and f (4) a, b, c, e and f

CGC264

Exercise-II ANSWER KEY

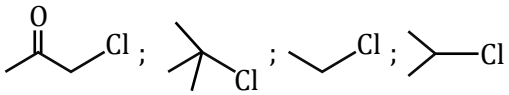
Question	1	2
Answer	**3**	**4**

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

Exercise - I — Haloalkanes and Haloarenes

1. The relative reactivity of following halides towards S_N2 reaction follows the order :-



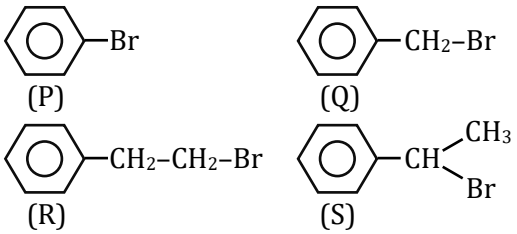
P Q R S

(1) Q > S > R > P (2) P > S > R > Q

(3) S > R > Q > P (4) P > R > S > Q

CHH205

2. Rate of S_N1 reaction is :-

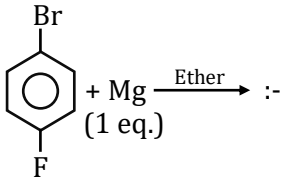


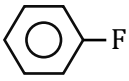
(P) (Q) (R) (S)

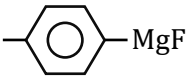
(1) S > Q > R > P (2) S > R > P > Q

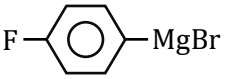
(3) P > Q > R > S (4) S > R > Q > P

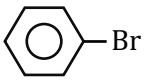
CHH206

3.  :-

(1) 

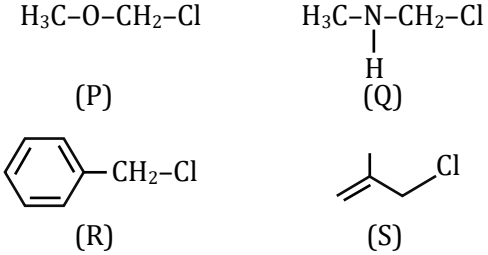
(2) 

(3) 

(4) 

CHH207

4. Reactivity order of S_N1 is :-



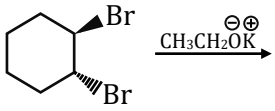
(P) (Q) (R) (S)

(1) P > Q > R > S (2) R > Q > P > S

(3) Q > P > R > S (4) Q > P > S > R

CHH208

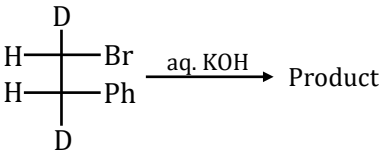
5. The most probable product in the following reaction is :-



(1)  (2) 

(3)  (4) 

CHH209

6. 

What is correct about product ?

(1) Diastereomeric pair

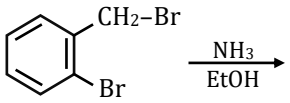
(2) Enantiomeric pair

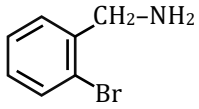
(3) Single stereoisomer

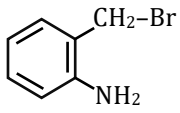
(4) Meso form of compound

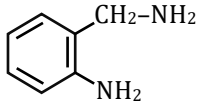
CHH210

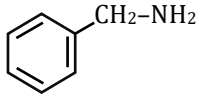
7. What is the major product obtained in the following reaction ?



(1) 

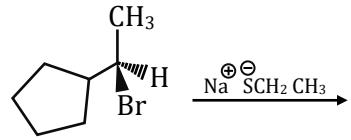
(2) 

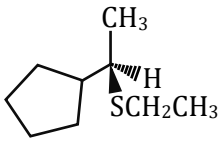
(3) 

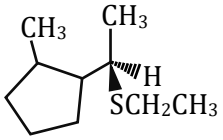
(4) 

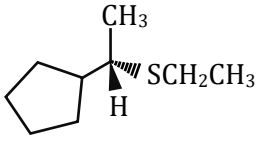
CHH211

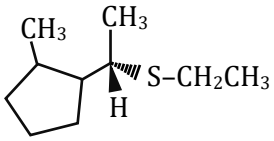
AIM : FREE EDUCATION TO ALL APUL

8.  Product is :-

(1) 

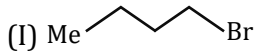
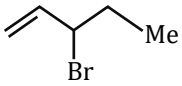
(2) 

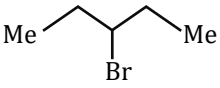
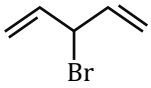
(3) 

(4) 

CHH212

9. Among the following alkyl bromide correct order of S_N1 reactivity is:

(I)  (II) 

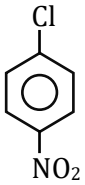
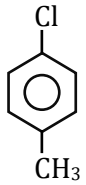
(III)  (IV) 

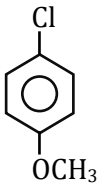
(1) I > II > III > IV (2) IV > II > III > I

(3) IV > III > II > I (4) II > IV > III > I

CHH213

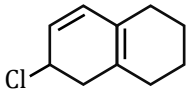
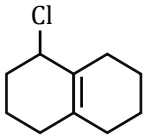
10. Which of the following compound undergoes hydrolysis most easily ?

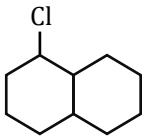
(1)  (2) 

(3)  (4) 

CHH214

11. The correct order for E2 reaction with alc. KOH will be:-

(i)  (ii) 

(iii) 

(1) i > ii > iii (2) iii > ii > i

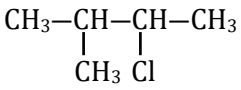
(3) ii > i > iii (4) iii > i > ii

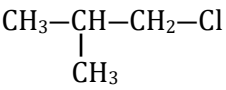
CHH215

12. Which of the following compound is most reactive toward S_N2 displacement?

(1) $CH_3CH_2CH_2CH_2–Cl$

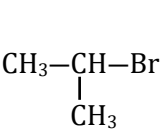
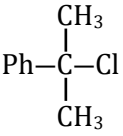
(2) $(CH_3)_3C–Cl$

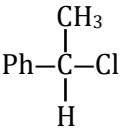
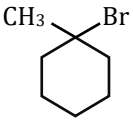
(3) 

(4) 

CHH216

13. Which one of the following will give racemised product in C_2H_5OH?

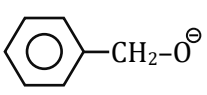
(1)  (2) 

(3)  (4) 

CHH217

14. Arrange the following group in order of their nucleophilic strength:-

(A) $CH_3–O^{\ominus}$ (B) $CH_3CH_2–S^{\ominus}$

(C) $CH_3–O–CH_2–O^{\ominus}$ (D) 

(1) A > B > C > D (2) D > B > C > A

(3) B > A > D > C (4) A > B > D > C

CHH218

AIM : FREE EDUCATION TO ALL APUL

15. Which of the following molecule would have a carbon halogen bond most susceptible to nucleophilic substitution?

(1) 2-Flourobutane (2) 2-Chlorobutane

(3) 2-Bromobutane (4) 2-Iodobutane

CHH219

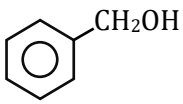
16. Which of the following compound is used as refrigerant?

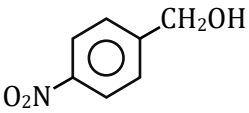
(1) Urea (2) Acrolein

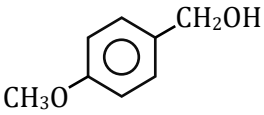
(3) CCl_2F_2 (4) $CH_2(OH)-CH_2(OH)$

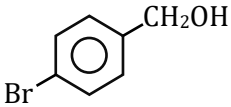
CHH220

17. Arrange the following compounds for their reactivity toward nucleophilic substitution with HBr?

(A) 

(B) 

(C) 

(D) 

(1) A > B > C > D (2) C > A > D > B

(3) D > C > B > A (4) B > D > A > C

CHH221

18. Which of the following reaction occur most rapidly?

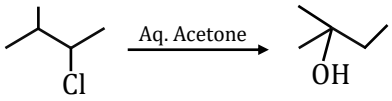
(1) $(CH_3)_2CHOH \xrightarrow{HBr}$

(2) $(CH_3)_2CHOH \xrightarrow{HI}$

(3) $(CH_3)_3C-OH \xrightarrow{HBr}$

(4) $(CH_3)_3C-OH \xrightarrow{HI}$

CHH222

19. 

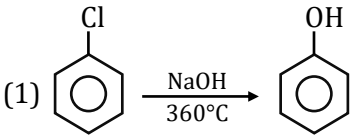
This reaction followed by :-

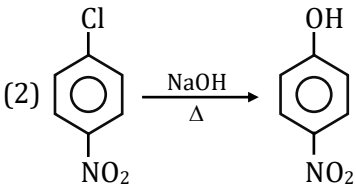
(1) S_N1 path (2) S_N2 path

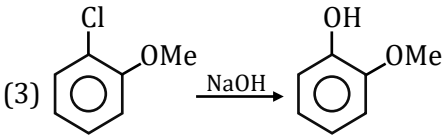
(3) E1 path (4) S_NAE path

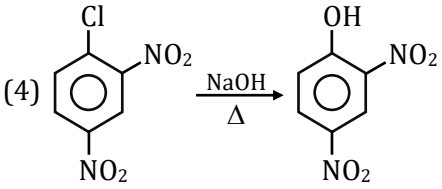
CHH223

20. Which reaction is least feasible ?

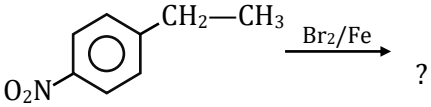
(1) 

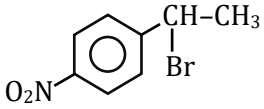
(2) 

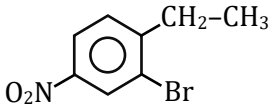
(3) 

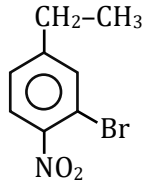
(4) 

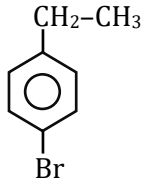
CHH224

21. 

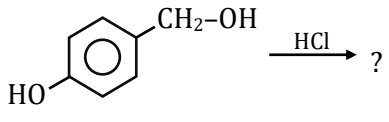
(1) 

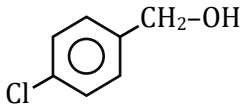
(2) 

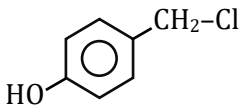
(3) 

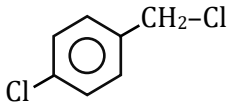
(4) 

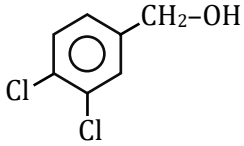
CHH225

22. 

(1) 

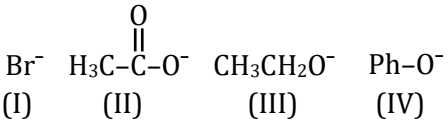
(2) 

(3) 

(4) 

CHH226

23. Arrange the following in order of their leaving group tendency ?



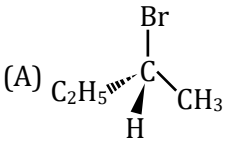
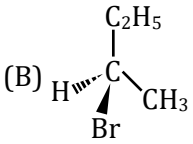
(I) (II) (III) (IV)

(1) I > II > III > IV (2) I >III > II > IV

(3) I > II > IV > III (4) I > IV > III > II

CHH227

24. The addition of HBr to 1-butene gives mixture of products A, B and C.

(A)  (B) 

(C) $CH_3–CH_2–CH_2–CH_2–Br$

The mixture consists of :-

(1) B as major, A and C as minor products

(2) A and B as major and C as major products

(3) B as major, A and C as major products

(4) A and B as major and C as minor products

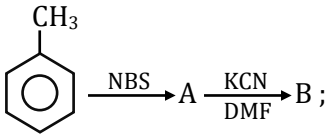
CHH228

Exercise-I ANSWER KEY

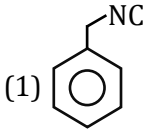
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**1**	**3**	**3**	**2**	**3**	**1**	**3**	**2**	**1**	**1**	**1**	**3**	**3**	**4**
Question	16	17	18	19	20	21	22	23	24						
Answer	**3**	**2**	**4**	**1**	**3**	**2**	**2**	**3**	**4**						

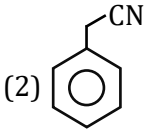
AIM : FREE EDUCATION TO ALL APUL

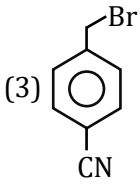
Exercise - II Haloalkanes and Haloarenes

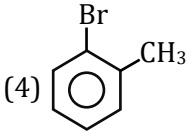
1. 

Product B is :-

(1) 

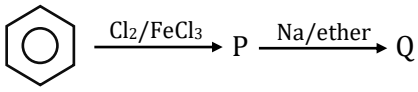
(2) 

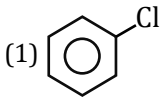
(3) 

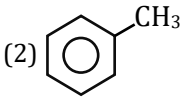
(4) 

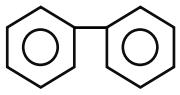
CHH229

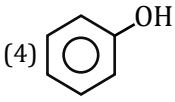
2. The end product (Q) in the following sequence of reaction is :-



(1) 

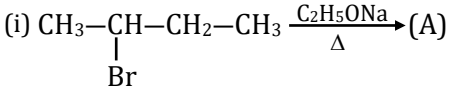
(2) 

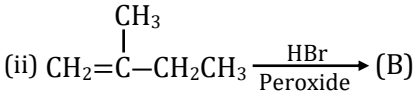
(3) 

(4) 

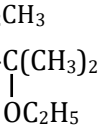
CHH230

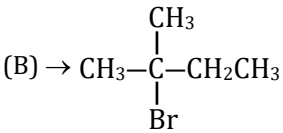
3. Identify the products (A) and (B).

(i) 

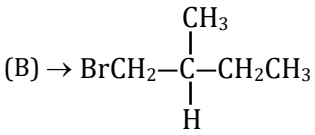
(ii) 

(1) (A) → $CH_3CH_2—CH_2—CH_3$,
(B) → $CH_3(CH_2)_2CH_3$

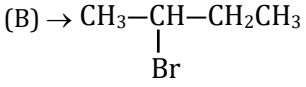
(2) (A) → 

(B) → 

(3) (A) → $CH_3—CH=CH—CH_3$,

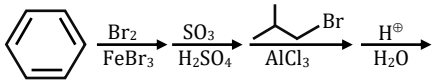
(B) → 

(4) (A) → $CH_3CH_2–CH_2CH_2CH_3$,

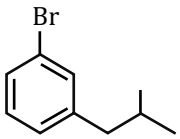
(B) → 

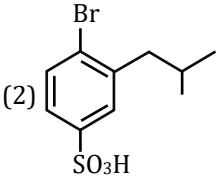
CHH231

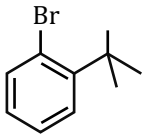
4. Give the major product from the following reaction sequence

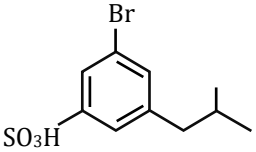


Major product

(1) 

(2) 

(3) 

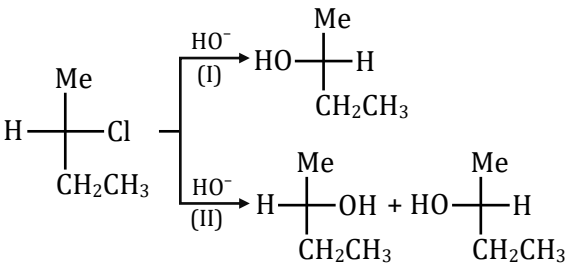
(4) 

CHH232

5. One mole of 1,2-dibromopropane on treatment with X moles of $NaNH_2$ followed by treatment with ethyl bromide give a pentyne. The value of X is :-

(1) One (2) Two (3) Three (4) Four

CHH233

6. 

Reaction I[st] and II[nd] are:

(1) Both S_N1 (2) Both S_N2

(3) (I) S_N1, (II) S_N2 (4) (I) S_N2, (II) S_N1

CHH234

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

7. Consider the reaction:

$CH_3–CH_2–CH_2–Br + NaCN \rightarrow$

$CH_3CH_2CH_2–C\equiv N + NaBr$

The **correct** statement is :-

(1) The reaction will be fastest in water

(2) The reaction will be fastest in N, N–dimethylformamide (DMF)

(3) Transition state of S_N2 is tetrahedral and sp^3 hybridized

(4) If conc. of alkyl bromide is tripled and conc. of $\overset{\ominus}{CN}$ is reduced to half rate of S_N2 increased by 2 times

CHH235

8. Which of the following statement is incorrect:-

(1) Boiling point of alkylhalide is R–I > R–Br > R–Cl > R–F

(2) Alkyl halide are soluble in organic solvent and less soluble in water

(3) Freons are used for aerosol propellant, refrigeration and air conditioning purpose.

(4) o and m - dichlorobenzene has higher melting point than those of p-isomer

CHH236

9. Which of the following is correctly match:-

(A) Chloramphenicol → Iodine containing Hormone

(B) Thyroxine → Chlorine containing antibiotic

(C) Chloroquine→ Treatment for malaria

(D) Fluorinated compound → Potential blood substitute

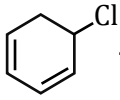
(E) Iodoform → Antiseptic

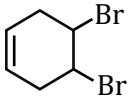
(1) A, B, C (2) B, C

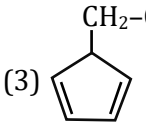
(3) C, D, E (4) D, E

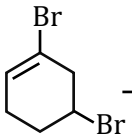
CHH237

10. In which of the following reaction benzene is not form as major product ?

(1)  $\xrightarrow[\Delta]{NaOH(alc.)}$ Product

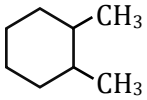
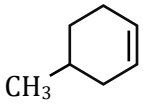
(2)  $\xrightarrow[\Delta]{KOH(alc.)}$ Product

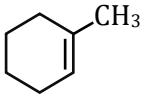
(3)  $\xrightarrow[\Delta]{H^{\oplus}}$ Product

(4)  $\xrightarrow[\Delta]{Zn(dust)}$ Product

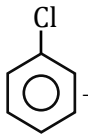
CHH238

11.  $+ Br_2 \xrightarrow{h\nu} A \xrightarrow[\Delta]{CH_3Na} B$; 'B' is–

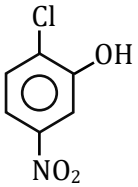
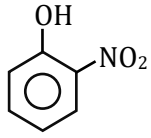
(1)  (2) 

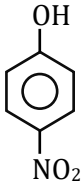
(3)  (4) 

CHH239

12.  $\xrightarrow[conc\,H_2SO_4]{conc\,HNO_3} A \xrightarrow[(ii)\ H^{\oplus}]{(i)\ NaOH/443K} B$

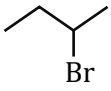
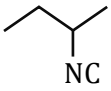
B as a major product is–

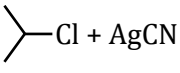
(1)  (2) 

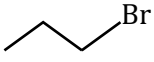
(3)  (4) 

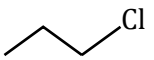
CHH240

13. Which is correct reaction

(1)  $\xrightarrow{KCN}$ 

(2) 

(3) 

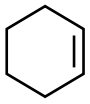
(4) 

CHH241

14. How many chiral compounds are possible on mono-chlorination of 2-Methyl butane?

(1) 6
(2) 8
(3) 2
(4) 4

CHH242

15.  $+ Cl_2 \xrightarrow{CCl_4}$ A + B, A and B are

(1) Enantiomers
(2) Diastereomers
(3) Identical
(4) Conformations

CHH243

AIM : FREE EDUCATION TO ALL APUL

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**3**	**3**	**3**	**3**	**4**	**2**	**4**	**3**	**4**	**4**	**3**	**4**	**4**	**1**

Exercise - I

Alcohols, Phenols and Ethers

1. An ether is more volatile then alcohol having the same molecular formula this is due to:-
(1) Inter molecular H–bonding in ether
(2) Inter molecular H-bonding in alcohols
(3) Dipolar character of ether
(4) Alcohol has resonating structure

CAE162

2. Among the following compounds which can be dehydrated most easily is:-
(1) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-OH$
(2) $CH_3-CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CH_3$
(3) $CH_3-CH_2-\underset{\displaystyle OH}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-CH_2-CH_3$
(4) $CH_3-CH_2-CH_2-CH_2-CH_2-OH$

CAE163

3. When CH_3-CH_2-COOH is reduced with $LiAlH_4$, the compound obtained will be:-
(1) $CH_3-CH_2-CH_3$
(2) $CH_2=CH-CH_2-OH$
(3) $CH_3-CH_2-CH_2-OH$
(4) CH_3-CH_2-CHO

CAE164

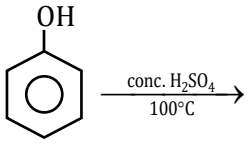
4. In the given reaction
$CH_3-CH=CH-CH_2-OH \xrightarrow{PCC}$ Product
Product is :
(1) $CH_3-CH_2-CH_2-CH_2-OH$
(2) $CH_3-CH_2-CH_2-CHO$
(3) $CH_3-CH=CH-CHO$
(4) $CH_3-CH=CH-COOH$

CAE165

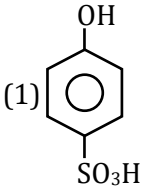
5. Reaction of phenol with dil. HNO_3 gives :
(1) p- and m-nitrophenol
(2) o- and p-nitrophenol
(3) Picric acid
(4) o- and m-nitrophenol

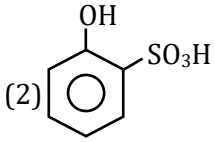
CAE166

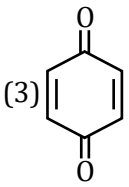
6. In following reaction sequence :

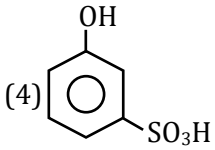


Major product is :

(1) 

(2) 

(3) 

(4) 

CAE167

7. In the given reaction
$C_6H_5-O-CH_2-CH_3 \xrightarrow{\text{Excess HI}/\Delta}$ [X] + [Y]
[X] and [Y] will respectively be :
(1) C_6H_5I and CH_3CH_2I
(2) C_6H_5OH and CH_3-CH_2-I
(3) C_6H_5I and CH_3CH_2OH
(4) C_6H_5OH and $CH_2=CH_2$

CAE168

8. In the given reaction
$CH_3-CH_2-CH_2-O-CH_2-CH_3 \xrightarrow{HI}$ [X] & [Y]
[X] and [Y] respectively be ;
(1) $CH_3CH_2CH_2OH$ and CH_3CH_2I
(2) $CH_3CH_2CH_2I$ and CH_3CH_2OH
(3) $CH_3CH_2CH_2I$ and $CH_2=CH_2$
(4) $CH_3-CH=CH_2$ and $CH_2=CH_2$

CAE169

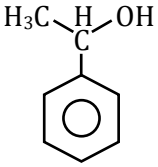
9. Amongst the following alcohols which would react fastest with conc. HCl and $ZnCl_2$?
(1) 2-Methylbutan-1-ol
(2) 2-Pentanol
(3) 1-Pentanol
(4) 2-Methyl-2-butanol

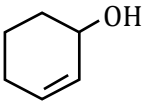
CAE170

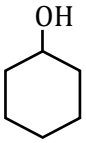
AIM : FREE EDUCATION TO ALL APUL

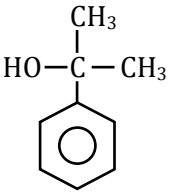
AIM : FREE EDUCATION TO ALL APUL

10. Minimum reactive towards dehydration in acidic medium

(1) 

(2) 

(3) 

(4) 

CAE171

11. Order of boiling points is–

(a) Pentan-1-ol (b) butane

(c) pentanal (d) ethoxyethane

(1) b > d > c > a (2) a > c > d > b

(3) c > d > b > a (4) d > a > b > c

CAE172

12. $CH_3CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-O-CH_2CH_3 + HI \rightarrow ?$

(1) $CH_3CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2OH + CH_3CH_2I$

(2) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2I + CH_3CH_2OH$

(3) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2I + CH_3-I$

(4) $CH_3—CH_2—OH$ only

CAE173

13. $CH_3—Br + NaO—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—CH_3 \longrightarrow ?$

(1) $CH_3—O—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—CH_3$

(2) $CH_3—\overset{\displaystyle CH_2}{\overset{\|}{\underset{|}{C}}}—\underset{\displaystyle CH_3}{\underset{|}{}}$

(3) $CH_2 = CH_2$

(4) $CH_3—O—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_3$

CAE174

14. The product of the following reaction is :-

$(CH_3)_3C-CHOH-CH_3 \xrightarrow[ZnCl_2]{Conc.HCl}$

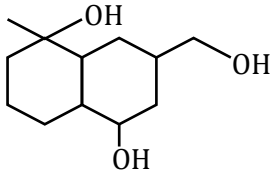
(1) $(CH_3)_3C-CHCl-CH_3$

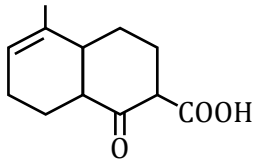
(2) $(CH_3)_3C-CH_2Cl$

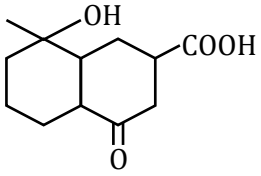
(3) $(CH_3)_2CCl-CH(CH_3)_2$

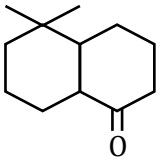
(4) $(CH_3)_2C=C(CH_3)_2$

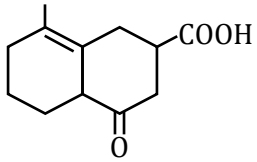
CAE175

15.  $\xrightarrow{KMnO_4/H^{\oplus}}$ Product :-

(1) 

(2) 

(3) 

(4) 

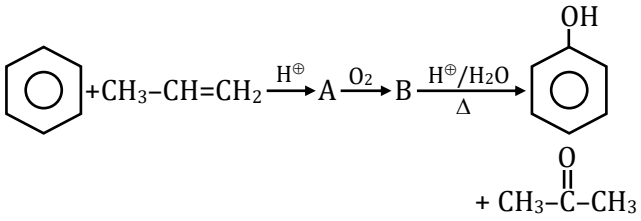
CAE176

Exercise-I ANSWER KEY

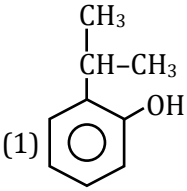
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	3	2	1	2	1	4	3	2	1	1	3	2

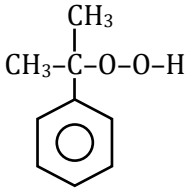
Exercise - II
Alcohols, Phenols and Ethers

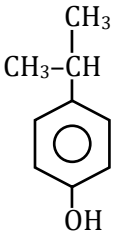
1. In following reaction

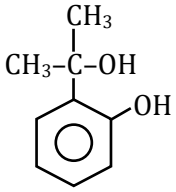


The structure of 'B' is:-

(1) 

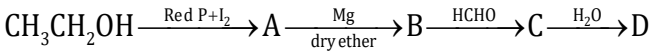
(2) 

(3) 

(4) 

CAE177

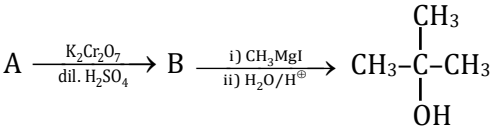
2. In the following sequence of reaction:



the compound D is :-

(1) Butanal (2) n-butyl alcohol
(3) n-propyl alcohol (4) Propanal

CAE178

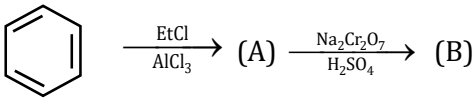
3. 

The reactant A is :

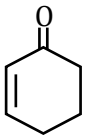
(1) $CH_3CHOHCH_3$ (2) CH_3CH_2CHO
(3) C_2H_5OH (4) CH_3COOH

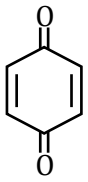
CAE179

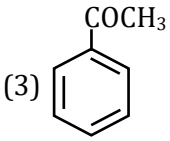
4. In the given reaction:

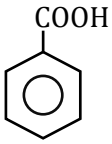


Product (B) is :

(1) 

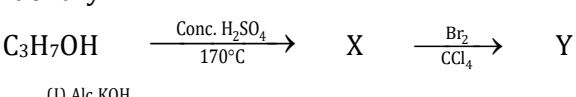
(2) 

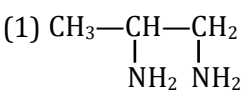
(3) 

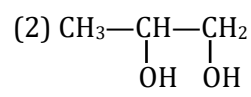
(4) 

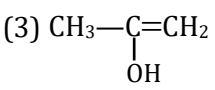
CAE180

5. In the following series of chemical reaction. Identify Z :



(1) 

(2) 

(3) 

(4) $CH_3-C\equiv CH$

CAE181

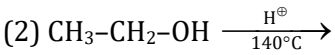
6. Methanol and ethanol are distinguished by the :

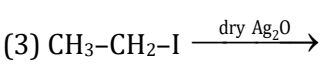
(1) Action of HCl
(2) Iodoform test
(3) Lucas test
(4) Sodium

CAE182

7. Which of the following give ether ?

(1) CH_3CH_2ONa and C_2H_5I

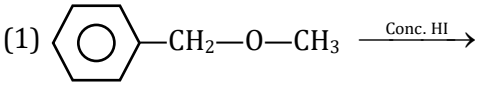
(2) 

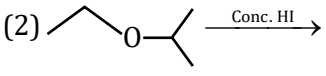
(3) 

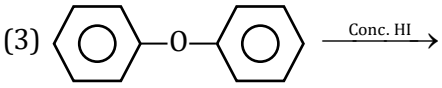
(4) All of the above

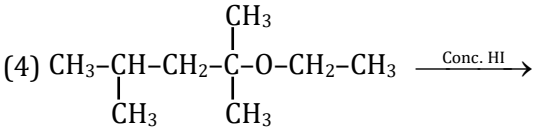
CAE183

8. Which of the following reaction followed by S_N2 path.

(1) 

(2) 

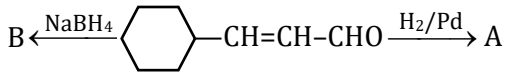
(3) 

(4) 

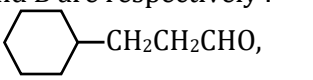
CAE184

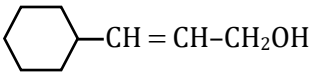
AIM : FREE EDUCATION TO ALL APUL

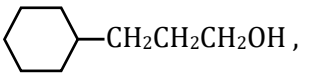
AIM : FREE EDUCATION TO ALL APUL

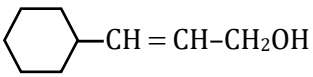
9. 

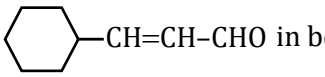
A and B are respectively :-

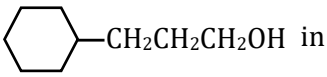
(1) 



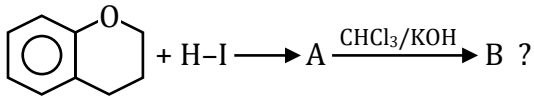
(2) 

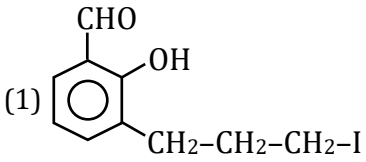


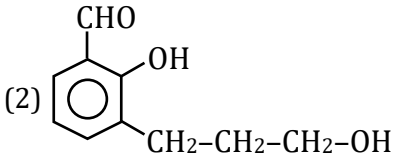
(3) 

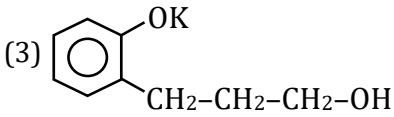
(4) 

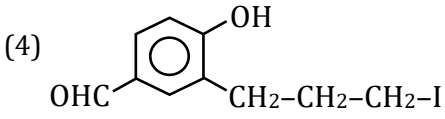
CAE185

10. 



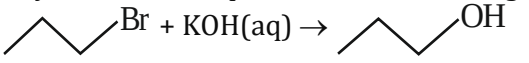




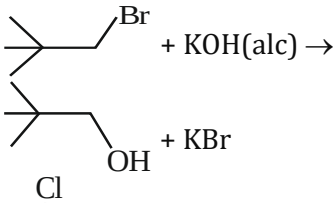


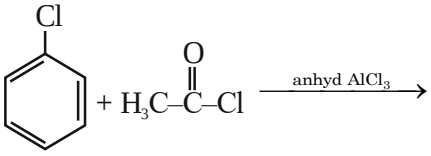
CAE186

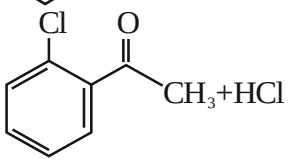
11. Identify the incorrect option from the following:

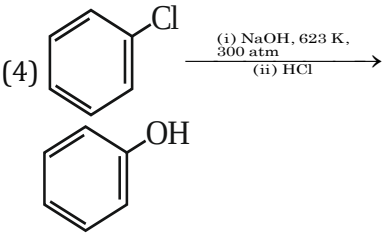
(1) 

+ KBr

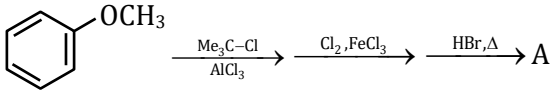
(2) 

(3) 

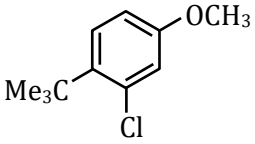


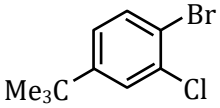
(4) 

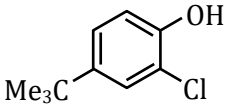
CAE187

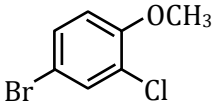
12. 

The final product A is :

(1) 

(2) 

(3) 

(4) 

CAE188

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	2	3	1	4	4	2	4	2	2	2	2	3

Exercise - I Aldehydes, Ketones and Carboxylic acids

1. Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is :-

(1) $CH_3-CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CH_3$

(2) $C_6H_5-CH_2-CH_2-OH$

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2OH$

(4) $PhCHOHCH_3$

CAA230

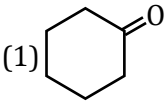
2. Butan-2-one is obtained, if following compound is reacted with methyl magnesium bromide followed by hydrolysis ?

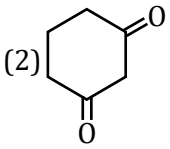
(1) CH_3-CHO (2) $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-Cl$

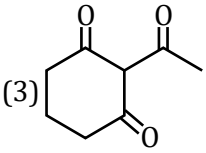
(3) $CH_3-C\equiv N$ (4) $CH_3-CH_2-C\equiv N$

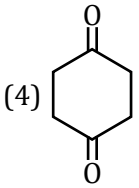
CAA231

3. Which of the compound have most acidic αH:

(1) 

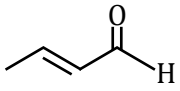
(2) 

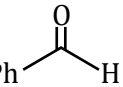
(3) 

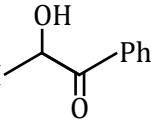
(4) 

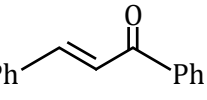
CAA232

4. Which of the following compounds does not give positive Tollens' test :

(1) 

(2) 

(3) 

(4) 

CAA233

5. Arrange the following in order of their reactivity towards CH_3MgBr?

(I) $CCl_3-\overset{\displaystyle O}{\overset{\|}{C}}-H$ (II) $H_3C-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3$

(III) $H_3C-\overset{\displaystyle O}{\overset{\|}{C}}-H$ (IV) $Ph-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3$

(1) I > IV > III > II (2) II > I > III > IV

(3) I > II > III >IV (4) I > III > II > IV

CAA234

6. $\underset{\displaystyle CHO}{\underset{|}{CHO}} \xrightarrow{NaOH}$ major product

(1) $\underset{\displaystyle CH_2-OH}{\underset{|}{CH_2-OH}}$ (2) $\underset{\displaystyle COONa}{\underset{|}{COONa}}$

(3) $\underset{\displaystyle COONa}{\underset{|}{CH_2-OH}}$ (4) $\underset{\displaystyle CHO}{\underset{|}{CH_2-OH}}$

CAA235

7. How many aldol product will be obtained in the following reaction (Structural only)

$CH_3CHO + PhCHO \xrightarrow{NaOH}$

(1) 2 (2) 3 (3) 4 (4) 1

CAA236

8. Give electrophilicity order of following carbonyl compounds

(I) Benzaldehyde

(II) p-Toluenealdehyde

(III) p-Nitrobenzaldehyde

(IV) Acetophenone

(1) I > IV > III > II (2) III > I > II > IV

(3) IV > II > I > III (4) II > IV > I > III

CAA237

9 Carboxylic carbon of acid is.....than carbonyl carbon:

(1) Less electrophilic

(2) More electrophilic

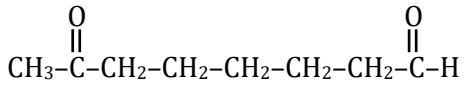
(3) less nucleophilic

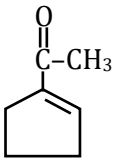
(4) More nucleophilic

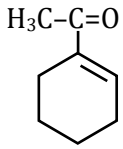
CAA238

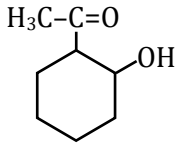
AIM : FREE EDUCATION TO ALL APUL

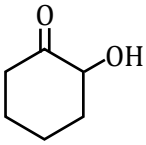
AIM : FREE EDUCATION TO ALL APUL

10. 
$\xrightarrow[\Delta]{\text{dil. NaOH}}$ (A) product is

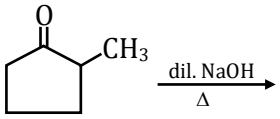
(1) 

(2) 

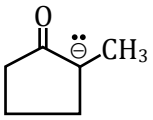
(3) 

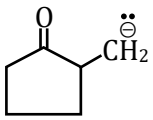
(4) 

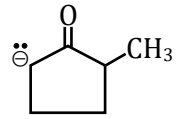
CAA239

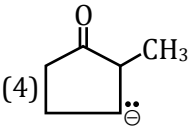
11. 

In the above Aldol condensation, attacking nucleophile will be ?

(1) 

(2) 

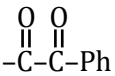
(3) 

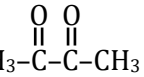
(4) 

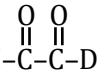
CAA240

12. Which of the following cannot undergo Intramolecular Cannizzaro Reaction ?

(1) 

(2) 

(3) 

(4) 

CAA241

13. In which of the following group, each member gives positive iodoform test ?

(1) methanol, ethanol, propanone

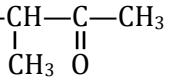
(2) ethanol, propanal, methanal

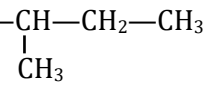
(3) ethanal, ethanol, isopropyl alcohol

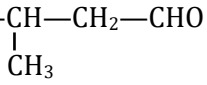
(4) propanal, 2-propanol, propanone

CAA242

14. Which gives isopentane on clemmensen reduction ?

(1) 

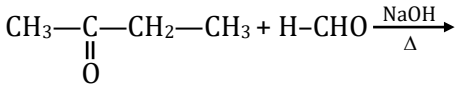
(2) 

(3) 

(4) All of them

CAA243

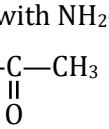
15. Number of structural products possible in following reaction are

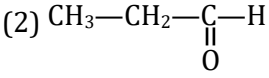


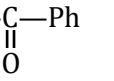
(1) 1 (2) 2 (3) 3 (4) 4

CAA244

16. The compound which gives stereo isomers on reaction with $NH_2–NH_2$?

(1) 

(2) 

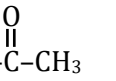
(3) 

(4) H–CHO

CAA245

17. Which of the following not shows stereoisomerism after aldol condensation.

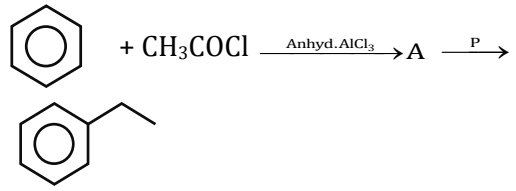
(1) $CH_3–CHO$

(2) 

(3) $CH_3–CH_2–CHO$

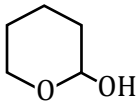
(4) $Ph–CH_2–CHO$

CAA246

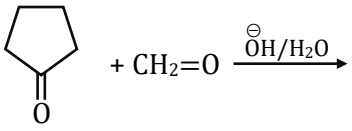
18. 

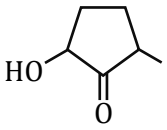
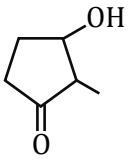
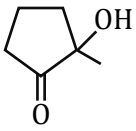
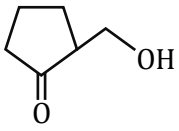
Reagent P can be
(1) HI/Red P
(2) Zn-Hg/HCl
(3) $NH_2-NH_2/\overset{\ominus}{O}H$
(4) All of these

CAA247

19. Which of the following compound will not give positive Tollens' test :-
(1) CH_3CHO
(2) 
(3) $CH_3CH_2CH_2CH_2\underset{\underset{OH}{|}}{CH}-OCH_3$
(4) $CH_3-CH(OCH_3)_2$

CAA248

20. 

The major product which can be isolated from this reaction is:
(1) 
(2) 
(3) 
(4) 

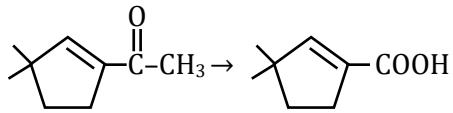
CAA249

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**4**	**3**	**4**	**4**	**3**	**1**	**2**	**1**	**2**	**3**	**3**	**3**	**4**	**4**
Question	16	17	18	19	20										
Answer	**2**	**2**	**4**	**4**	**4**										

AIM : FREE EDUCATION TO ALL APUL

Exercise - II Aldehydes, Ketones and Carboxylic acids

1. 

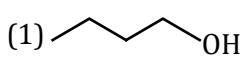
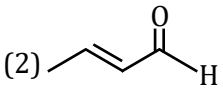
Suitable reagent for above conversion is

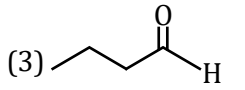
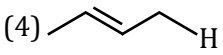
(1) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ \text{Tollen's Reagent}}$ (2) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ \text{Benzoyl peroxide}}$

(3) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ I_2/NaOH}$ (4) $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

CAA250

2. But-2-ene $\xrightarrow[2.Zn,H_2O]{1.O_3}$ P $\xrightarrow[\Delta]{OH^{\ominus}}$ Q $\xrightarrow[H_2(1eq)]{Pd/C}$ R is :

(1)  (2) 

(3)  (4) 

CAA251

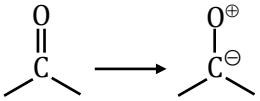
3. Which of the following statement is not true about carbonyl group :

(1) Carbon-oxygen bond is polarised due to higher electronegativity of oxygen

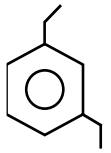
(2) Carbonyl carbon is electrophilic while oxygen is nucleophilic center

(3) Carbonyl group have more dipole moment than ether

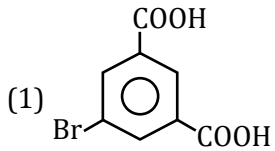
(4) Polarity of carbonyl group as given

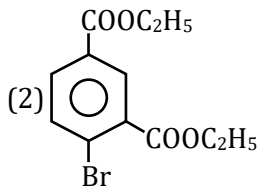


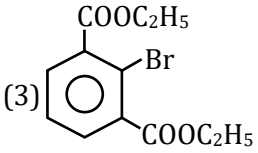
CAA252

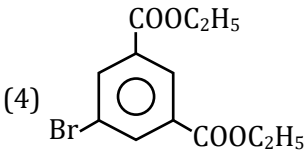
4.  $\xrightarrow{KMnO_4/H^+}$ A $\xrightarrow[FeBr_3]{Br_2}$ B $\xrightarrow[H^{\oplus}]{C_2H_5OH}$ C

C is :

(1) 

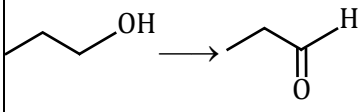
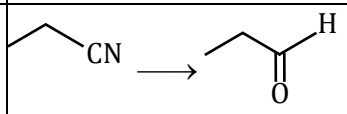
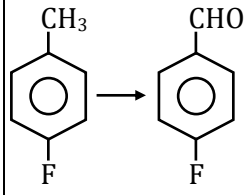
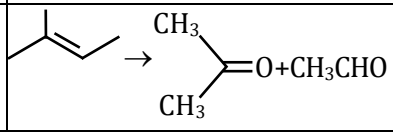
(2) 

(3) 

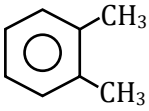
(4) 

CAA253

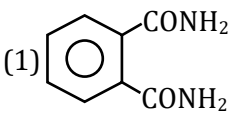
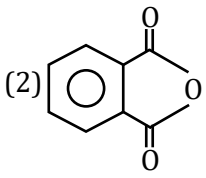
5. Which of the following is not correctly matched:

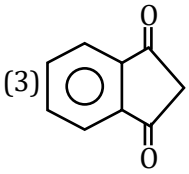
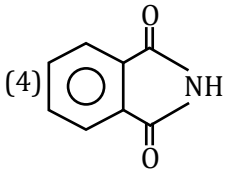
	Reaction	Reagent
1.		PCC
2.		DIBAL-H
3.		$CrO_3 + Ac_2O$ /H_3O^+
4.		$KMnO_4/H^+$

CAA254

6.  $\xrightarrow[\Delta]{KMnO_4/H^+}$ A $\xrightarrow[\text{Strong }\Delta]{NH_3}$ B

B is :

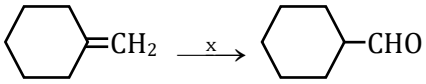
(1)  (2) 

(3)  (4) 

CAA255

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

7. 

reagent X is :

(1) (a) BH_3, THF (b) H_2O_2 / OH^-

(2) H_3O^+

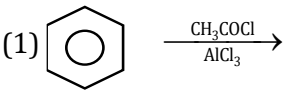
(3) (a) O_3 (b) Zn/H_2O

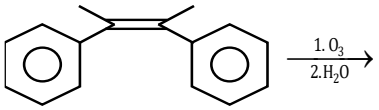
(4) (a) (i) BH_3/THF (ii) H_2O_2/OH^-

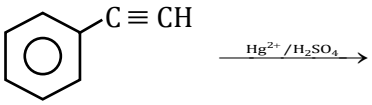
(b) Anhydrous CrO_3

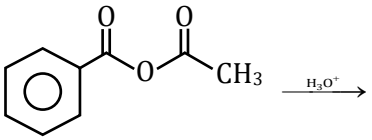
CAA256

8. Acetophenone is not produced in which reaction:

(1) 

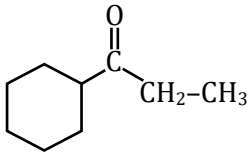
(2) 

(3) 

(4) 

CAA257

9. Which of the following compound can form hydrazone and will give iodoform test but will not give Tollens' test?

(1) 

(2) $H_3C-\underset{\underset{O}{\|}}{C}-CH_2-CH_3$

(3) $H_3C-\underset{\underset{O}{\|}}{C}-CH_2-CHO$

(4) CH_3-CH_2-OH

CAA258

10. Acetaldehyde and benzaldehyde cannot be distinguished by

(1) Tollens' reagent (2) Fehling solution

(3) Benedict solution (4) Iodoform test

CAA259

11. Benzaldehyde and acetaldehyde can be distinguish by :

(1) Hinsberg reagent (2) Tollens' reagent

(3) Baeyer's reagent (4) Iodoform test

CAA260

12. In which reaction racemic mixture is obtained as a product?

(1) $Ph-CHO+HCN \rightarrow$

(2) $H_3C-\underset{\underset{C_2H_5}{|}}{CH}-CH_2-OH + SOCl_2 \xrightarrow{\text{Pyridine}}$

(3) $H_3C-\underset{\underset{Cl}{|}}{CH}-CH_2-CH_3 + C_3H_7\overset{\ominus\oplus}{O}Na \xrightarrow[\Delta]{ROH}$

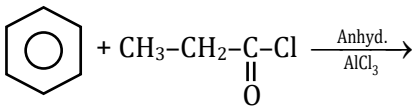
(4) All of these

CAA261

13. In which reaction ethyl phenyl ketone is obtained

(1) $C_6H_5-C\equiv N + Et\,MgBr \rightarrow A \xrightarrow[H^\oplus]{H_2O}$

(2) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-Cl + Ph_2Cd \rightarrow$

(3) 

(4) All of them

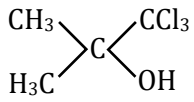
CAA262

14. $Ph-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow{NaOCl} A + salt$

$A + CH_3-\underset{\underset{O}{\|}}{C}-CH_3 \rightarrow B$, 'B' is

(1) $CH_3-\underset{\underset{O}{\|}}{C}-CH=\underset{\underset{CH_3}{|}}{C}-Ph$

(2) $Ph-\underset{\underset{O}{\|}}{C}-CH=CH-\underset{\underset{O}{\|}}{C}-CH_3$

(3) 

(4) $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3$

CAA263

AIM : FREE EDUCATION TO ALL APUL

15. $CH_3CH_2COOH + Cl_2 \xrightarrow{\text{Red P}} A \xrightarrow[\Delta]{NaOH/CaO} B \xrightarrow[\Delta]{\text{Alc. KOH}}$ C, 'C' is

(1) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$ (2) $CH_3-CH=CH_2$

(3) $CH_3-CH_2-\underset{\underset{O}{||}}{C}-OR$ (4) $CH_2=CH_2$

CAA264

16. Ethyl amine and Aniline can be distinguished by

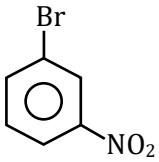
(1) By Isocyaninde test

(2) Azo dye test

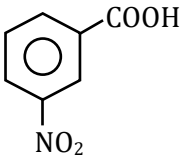
(3) Hinsberg's reagent

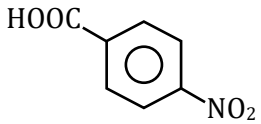
(4) Grignard reagent

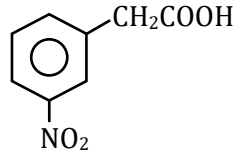
CAA265

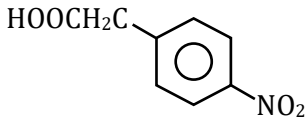
17.  $\xrightarrow[\text{dry ether}]{Mg} P \xrightarrow[\text{(dry ice)}]{CO_2} Q \xrightarrow{H_3\overset{\oplus}{O}}$;

R is–

(1) 

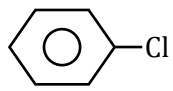
(2) 

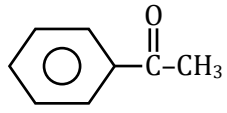
(3) 

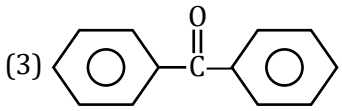
(4) 

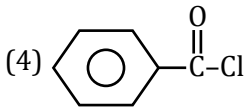
CAA266

18. $H_3C-COOH \xrightarrow[\Delta]{P_2O_5} A \xrightarrow{PCl_5} B \xrightarrow[AlCl_3]{C_6H_6} C$; C is–

(1) 

(2) 

(3) 

(4) 

CAA267

19. Which of the following is not correctly matched

Compound	Use
(1) Ethanoic acid	Solvent
(2) Ester of benzoic acid	Perfumes
(3) Sodium benzoate	Preservatives
(4) Iodoform	Anesthetic

CAA268

20. Identify the end product:

$CH_3CH_2CH_2CHO \xrightarrow[(ii)\ H^{\oplus}]{\text{(i) Ammonical } AgNO_3}$

$X \xrightarrow{NaHCO_3} Y \xrightarrow[\Delta]{(NaOH+CaO)} Z$

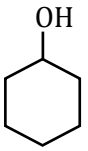
(1) $CH_3-CH_2CH_2-COOH$

(2) $CH_3-CH_2-CH_2-CH_3$

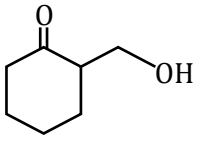
(3) CH_3-CH_2-COOH

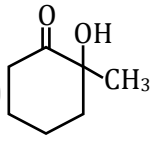
(4) $CH_3-CH_2-CH_3$

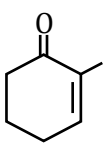
CAA269

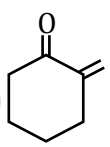
21.  $\xrightarrow{PCC}$ (A) $\xrightarrow[dil.KOH/\Delta]{HCHO}$ (B)

(B) is :

(1) 

(2) 

(3) 

(4) 

CAA270

AIM : FREE EDUCATION TO ALL APUL

22. Alkene (X) (C_5H_{10}) on ozonolysis gives a mixture of two compounds (Y) and (Z). Compound (Y) and (Z) gives positive Fehling's test, (Z) does not give idoform test. Compounds (X), (Y) and (Z) are :-

	X	Y	Z
(1)	$CH_2=C(CH_3)-CH_2CH_3$	HCHO	$CH_3-C(=O)-CH_2CH_3$
(2)	$CH_3-CH=C(CH_3)-CH_3$	CH_3CHO	CH_3COCH_3
(3)	$CH_3CH_2CH=CH_2$	CH_3CH_2CHO	HCHO
(4)	$CH_3CH=CHCH_2CH_3$	CH_3CHO	CH_3CH_2CHO

CAA271

23. In the following sequence of reaction, the final product (Z) is :

$$CH\equiv CH \xrightarrow{Hg^{2+}/dil.H_2SO_4} X \xrightarrow[(ii)\ H_2O]{(i)\ CH_3MgBr} Y \xrightarrow{[O]} Z$$

(1) Ethanal
(2) Propan-2-ol
(3) Propanone
(4) Propan-1-ol

CAA272

24. Identify the products (X) and (Y) in the given reaction:



$$\xrightarrow[AlCl_3]{(CH_3CO)_2O} X \xrightarrow[Conc.\ H_2SO_4]{Conc.\ HNO_3} Y$$

(1) X = Acetophenone,
Y = m-Nitroacetophenone
(2) X = Toluene,
Y = m-Nitroacetotoluene
(3) X = Acetophenone,
Y = o and p-Dinitroacetophenone
(4) X = Benzaldehyde,
Y = m-Nitrobenzaldehyde

CAA273

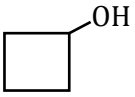
25. $CH_3-CH(OH)-CH_2-CH_2-C(=O)-CH_3 \longrightarrow CH_3-CH(OH)-(CH_2)_3-CH_3$

Above conversion can be achieved by :-
(1) Wolff-Kishner reduction
(2) Clemmensen reduction
(3) $LiAlH_4$
(4) $NaBH_4$

CAA274

26. Compound (X) C_4H_8O, which reacts with 2,4-DNP derivative but gives negative haloform test is :-

(1) $CH_3-C(=O)-CH_2-CH_3$ (2) $CH_3-CH(CH_3)-CHO$

(3)  (4) $CH_3-CH_2-CH(OH)-CH_3$

CAA275

27. $Ph-CH(OH)-CH_3 \xrightarrow{PCC} (A) \xrightarrow{NH_2-NH-C(=O)-NH_2} (B)$

Product (B) is :-

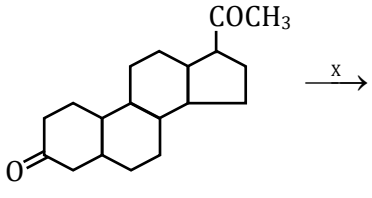
(1) $Ph-C(CH_3)=N-C(=O)-NH-NH_2$

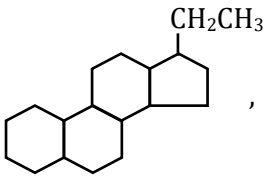
(2) $Ph-C(CH_3)=N-NH-C(=O)-NH_2$

(3) $Ph-CH=N-N(CH_3)-C(=O)-NH_2$

(4) $Ph-CH=N-C(=O)-NH_2$

CAA276

28. 

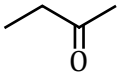


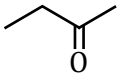
X can be
(1) NH_2-NH_2/KOH
(2) Zn–Hg/HCl
(3) Red P + HI
(4) All

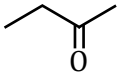
CAA277

AIM : FREE EDUCATION TO ALL APUL

29. Which of the following is correct ?

(1)  $\xrightarrow[HCl]{KCN}$ Diastereomeric pair

(2)  $\xrightarrow[H^{\oplus}]{NH_2-OH}$ Geometrical pair

(3)  $\xrightarrow[dry\ HCl]{MeOH(1\ eq)}$ Acetal

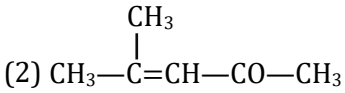
(4)  $\xrightarrow[dry\ HCl]{MeOH(2\ eq)}$ Racemic mixture

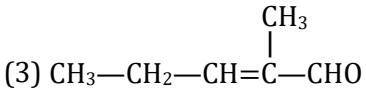
CAA278

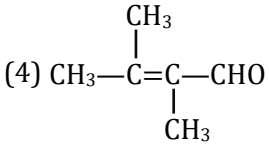
30. $CH_3-C\equiv CH \xrightarrow[THF]{BH_3} A \xrightarrow[^-OH]{H_2O_2} B \xrightarrow[\Delta]{\bar{O}H} C$

Product 'C' is

(1) $CH_3—CH_2—CH=CH—CH_2—CHO$

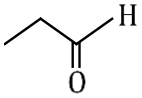
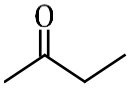
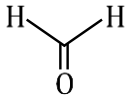
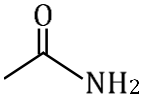
(2) 

(3) 

(4) 

CAA279

31. When Grignard reagent is reacted with a compound of Column-I, Product in Column-II is obtained

	Column-I		Column-II
(A)		(P)	3° Alcohol
(B)		(Q)	1° Alcohol
(C)		(R)	Alkane
(D)		(S)	2° Alcohol

Correct match is

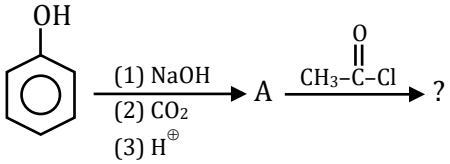
(1) A → S ; B → P ; C → Q ; D → P

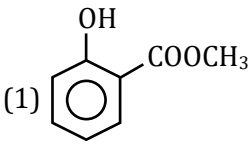
(2) A → S ; B → P ; C → Q ; D → R

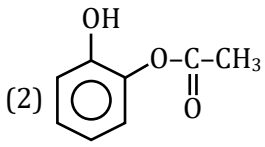
(3) A → S ; B → P ; C → R ; D → Q

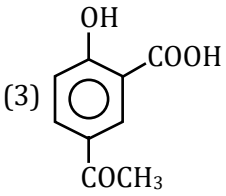
(4) A → Q ; B → P ; C → R ; D → R

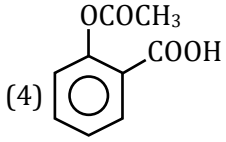
CAA280

32. 

(1) 

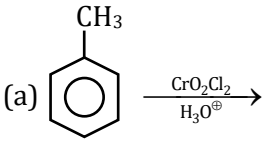
(2) 

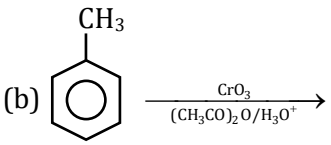
(3) 

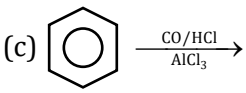
(4) 

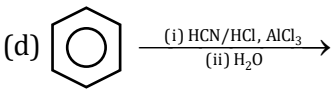
CAA281

33. Which of the following reaction form benzene carbaldehyde as a product :

(a) 

(b) 

(c) 

(d) 

(1) a, b, c

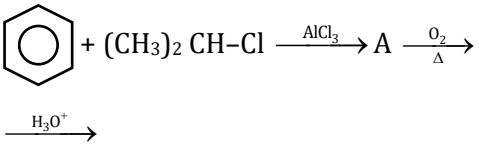
(2) a, b, d

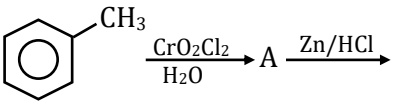
(3) a, b

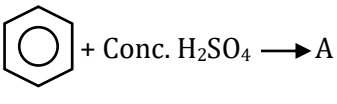
(4) a, b, c, d

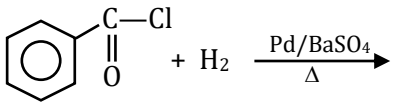
CAA282

34. Which is correct reaction regarding to phenol as major product ?

(1) 

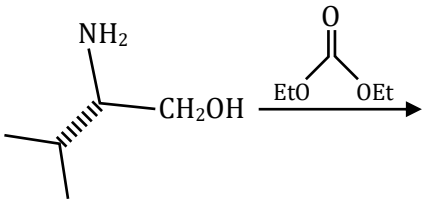
(2) 

(3) 

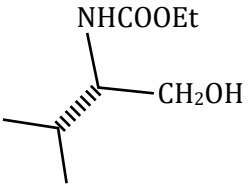
(4) 

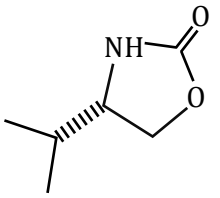
CAA283

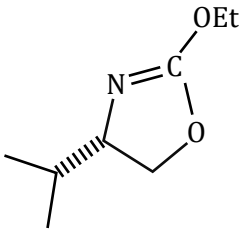
35. In the following reaction, 'A' is

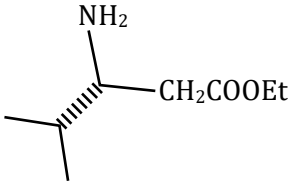


'A' Major product.

(1) 

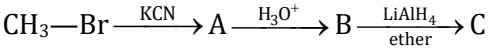
(2) 

(3) 

(4) 

CAA284

36. In the following sequence of reactions,



the end product C is :

(1) acetone

(2) methane

(3) acetaldehyde

(4) ethyl alcohol

CAA285

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	4	4	4	4	4	4	2	1	4	1	4	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	2	4	4	4	4	3	1	1	2	2	4	2	3
Question	31	32	33	34	35	36									
Answer	2	4	4	3	2	4									

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

Exercise - I Amines

1. Arrange the following in decreasing order of boiling point :

(I) $CH_3CH_2CH_2OH$ (II) $CH_3CH_2CH_2NH_2$

(III) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{|}}-N-CH_3$ (IV) $CH_3CH_2-NH-CH_3$

(V) $CH_3CH_2CH_2CH_3$

(1) I > II > IV > III > V

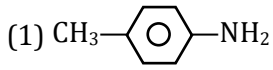
(2) III > IV > II > I > V

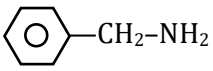
(3) I > V > II > III > IV

(4) II > I > III > IV > V

CAM157

2. Which of the following compound cannot be formed by Gabriel phthalimide synthesis ?

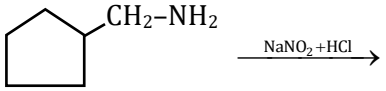
(1) 

(2) 

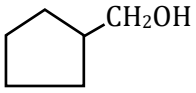
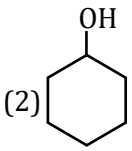
(3) $(CH_3)_2CH-NH_2$

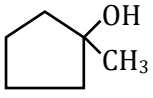
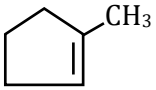
(4) $(C_2H_5)(CH_3)CH-NH_2$

CAM158

3. 

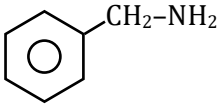
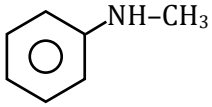
Find out major product :

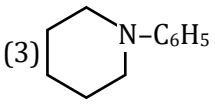
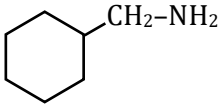
(1)  (2) 

(3)  (4) 

CAM159

4. Which of the following compound gives yellow oily liquid nitrosamine when reacted with nitrous acid:

(1)  (2) 

(3)  (4) 

CAM160

5. Given below are two statements :

Statement I :

$-NHCOCH_3$ group is also o/p-directing like $-NH_2$ group but it is less activating than $-NH_2$ group.

Statement II :

Lone pair of nitrogen in $-NHCOCH_3$ group is in resonance with carbonyl group. In the light of the above statements, choose the most appropriate answer from the options given below :

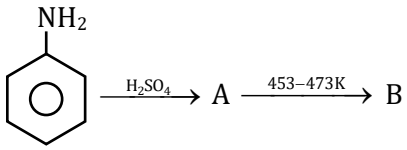
(1) Statement I is correct but statement II is incorrect.

(2) Both statement I and statement II are correct.

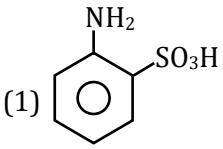
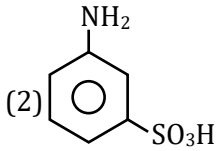
(3) Statement I is incorrect and statement II is correct.

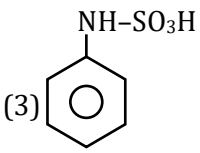
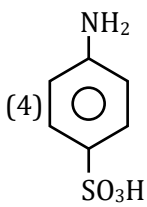
(4) Both statement I and statement II are incorrect.

CAM161

6. 

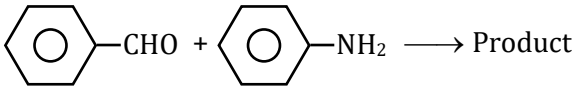
Compound B is :

(1)  (2) 

(3)  (4) 

CAM162

AIM : FREE EDUCATION TO ALL APUL

7.  Product

Product A is called :

(1) Azo dye (2) Schiff reagent

(3) Schiff base (4) Oxime

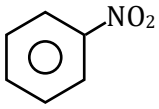
CAM163

8. Aryl fluoride may be prepared from arene diazonium chloride using

(1) CuF/HF (2) Cu/HF

(3) HBF_4/Δ (4) $HBF_4/NaNO_2$, Cu

CAM164

9.  Product is

(1) Phenyl hydroxylamine

(2) p-amino phenol

(3) Azoxybenzene

(4) Aniline

CAM165

10. $R-CN \xrightarrow[HCl]{SnCl_2} \xrightarrow{H_3O^{\oplus}} R-CHO$

Above reaction is

(1) Mendius reaction

(2) Stephen's reaction

(3) Rosenmund reaction

(4) Perkin reaction

CAM166

11. Which of the following has maximum Boiling Point ?

(1) Methoxyethane (2) n-Butane

(3) Propanal (4) Acetone

CAM167

12. Aniline cannot be prepared by Gabriel phthalimide synthesis due to–

(1) Bulky nature of benzene Ring

(2) Aryl halide do not undergo nucleophilic substitution reaction (S_N2).

(3) Aryl halide do not undergo electrophilic substitution reaction

(4) Both (1) & (3)

CAM168

13. Which one can give propanamine on reaction with $LiAlH_4$?

(1) Propyl cyanide

(2) N-Methylpropanamide

(3) Propanenitrile

(4) Both (1) & (3)

CAM169

14. 'A' $\xrightarrow{Br_2+NaOH}$ $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-NH_2$ Find out the structure of compound 'A'

(1) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2$

(2) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2$

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\overset{\displaystyle O}{\overset{\|}{C}}-OH$

(4) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2$

CAM170

15. Which one is soluble in NaOH :–

(1) N-Ethyl benzenesulphonamide

(2) N,N-diethyl benzenesulphonamide

(3) N-ethyl aniline

(4) N-methyl ethanamine

CAM171

16. Which of the following compound will give Hoffmann bromamide reaction ?

(1) $Ph-\overset{\displaystyle O}{\overset{\|}{C}}-NH-CH_3$

(2) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{N}}-CH_3$

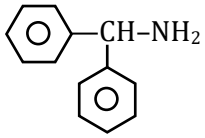
(3) $Ph-NH_2$

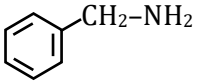
(4) $CH_3-CH_2-\overset{\displaystyle O}{\overset{\|}{C}}-NH_2$

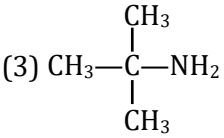
CAM172

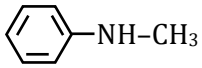
AIM : FREE EDUCATION TO ALL APUL

17. Which of the following compound will not give Carbylamine Reaction ?

(1) 

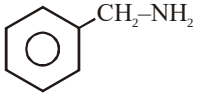
(2) 

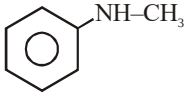
(3) 

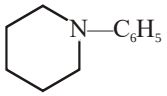
(4) 

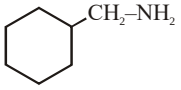
CAM173

18. Which of the following compound gives yellow oily liquid nitrosamine when react with nitrous acid:

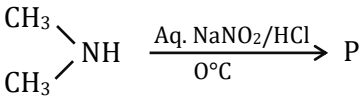
(1) 

(2) 

(3) 

(4) 

CAM174

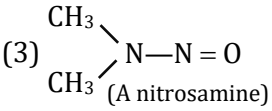
19. 

(2° Amine)

P should be

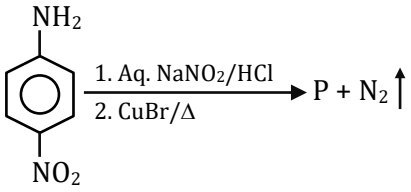
(1) CH_3—OH

(2) CH_3—Cl

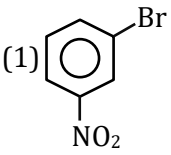
(3) 

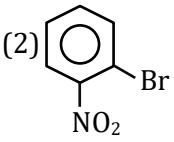
(4) All of these

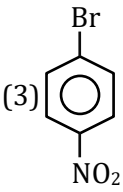
CAM175

20. 

P should be

(1) 

(2) 

(3) 

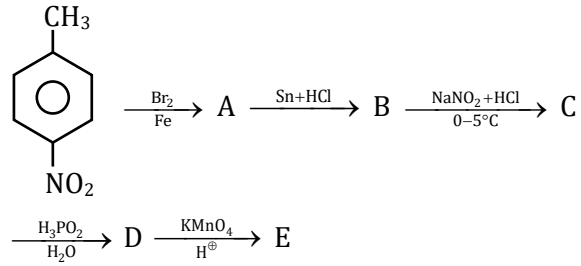
(4) A mixture of all of these

CAM176

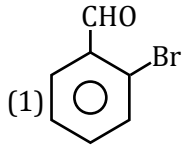
Exercise-I ANSWER KEY

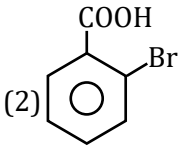
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**1**	**2**	**2**	**2**	**4**	**3**	**3**	**4**	**2**	**4**	**2**	**3**	**2**	**1**
Question	16	17	18	19	20										
Answer	**4**	**4**	**2**	**3**	**3**										

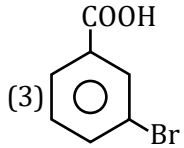
Exercise - II — Amines

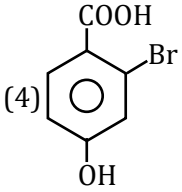
1. 

End product E will be :

(1) 

(2) 

(3) 

(4) 

CAM177

2. Which is Sandmeyer reaction among the following :

(1) $C_6H_5N_2Cl \xrightarrow[\text{or LAH}]{Na/C_2H_5OH} C_6H_5H$

(2) $C_6H_5N_2Cl \xrightarrow{Br_2/KOH} C_6H_5Br$

(3) $C_6H_5N_2Cl \xrightarrow[\Delta]{Cu_2Cl_2/HCl} C_6H_5Cl$

(4) $C_6H_5N_2Cl \xrightarrow[Cu/HCl]{} C_6H_5Cl$

CAM178

3. $CH_3Cl \xrightarrow{KCN} A \xrightarrow[\text{Hydrolysis}]{\text{Partial}} B \begin{cases} \xrightarrow{PCl_5} C \\ \xrightarrow{LAH} D \end{cases}$

Products C and D are respectively :

(1) CH_3COCl and CH_3CH_2OH

(2) CH_3COCl and $CH_3CH_2NH_2$

(3) CH_3CN and $CH_3CH_2NH_2$

(4) CH_3CH_2Cl and CH_3CH_2OH

CAM179

4. Which of the following conversion is incorrect:

(1) $CH_3CN \xrightarrow[\text{or LAH}]{Na/C_2H_5OH} CH_3CH_2NH_2$

(2) $CH_3CONH_2 \xrightarrow{Br_2/KOH} CH_3CH_2NH_2$

(3) $CH_3CONH_2 \xrightarrow[\Delta]{H^+/H_2O} CH_3COOH$

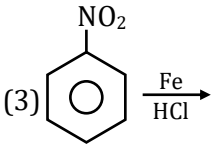
(4) $CH_3CONH_2 \xrightarrow{P_2O_5} CH_3CN$

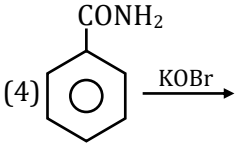
CAM180

5. Aniline cannot be obtained by :

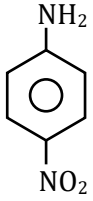
(1) $C_6H_5NO_2 \xrightarrow[\text{Electrolysis}]{\text{conc. } H_2SO_4}$

(2) $C_6H_5Cl \xrightarrow[\Delta]{NaNH_2}$

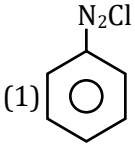
(3) 

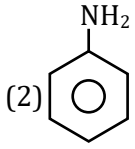
(4) 

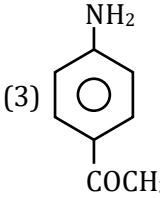
CAM181

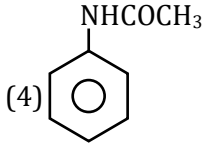
6. (A) $\xrightarrow{(CH_3CO)_2O}$ (B) $\xrightarrow[(ii)\ H_3O^+]{\text{(i)Nitrating mixture}}$ 

Find out (A) :

(1) 

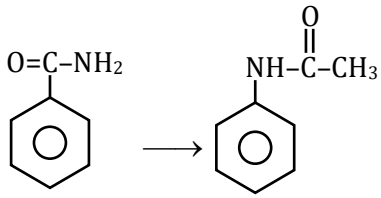
(2) 

(3) 

(4) 

CAM182

7. The reagents needed to convert :



(1) KOH, Br_2 ; $LiAlH_4$

(2) KOH, Br_2 ; CH_3COCl

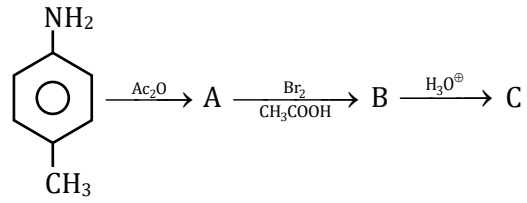
(3) KOH, Br_2 ; Ni,H_2 ; CH_3COCl

(4) HONO ; $(CH_3CO)_2O$

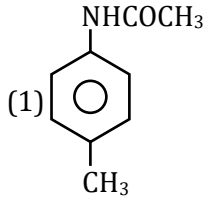
CAM183

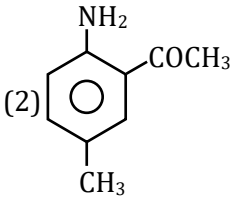
AIM : FREE EDUCATION TO ALL APUL

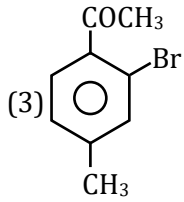
AIM : FREE EDUCATION TO ALL APUL

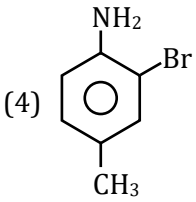
8. 

'C' would be :

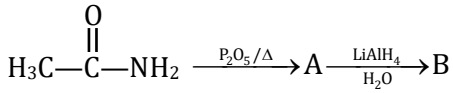
(1) 

(2) 

(3) 

(4) 

CAM184

9. 

Final product B is

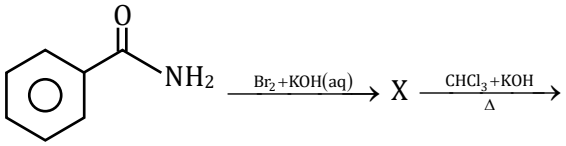
(1) A primary alcohol

(2) A primary amine

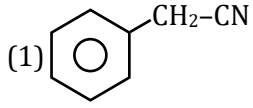
(3) A secondary amine

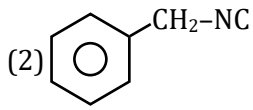
(4) A carboxylic acid

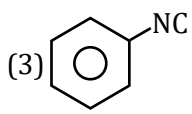
CAM185

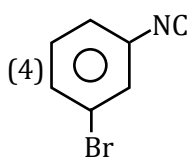
10. 

Y; Y is

(1) 

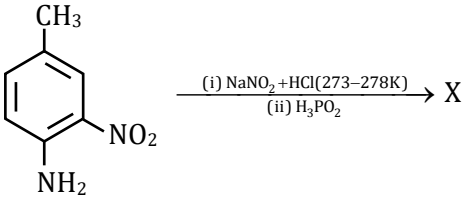
(2) 

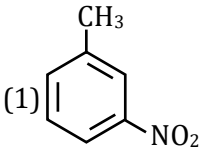
(3) 

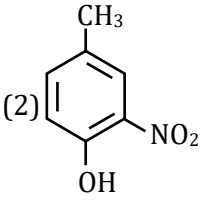
(4) 

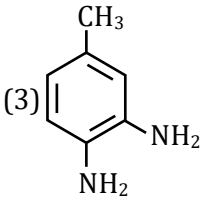
CAM186

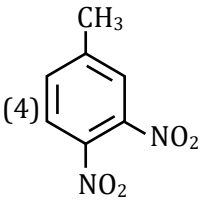
11. The structure of product X in the following reaction is :-



(1) 

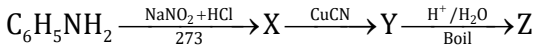
(2) 

(3) 

(4) 

CAM187

12. Identify 'Z' in the sequence :

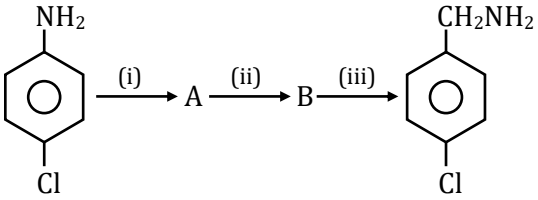


(1) C_6H_5CN (2) $C_6H_5CONH_2$

(3) C_6H_5COOH (4) $C_6H_5CH_2NH_2$

CAM188

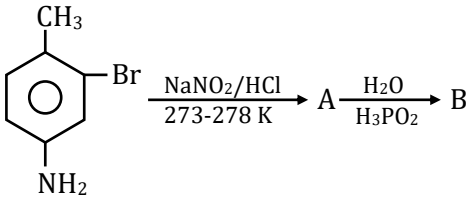
13. Mark the correct route of the conversion of p-chloroaniline to p-chlorobenzylamine.



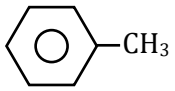
	(i)	(ii)	(iii)
(1)	Alkylation	KCN	H_2/Pt
(2)	Diazotisation	CuCN	H_2/Pt
(3)	Oxidation	H_2/Pt	Hydrolysis
(4)	Diazotisation	H_2O/H^+	Sn/HCl

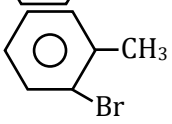
CAM189

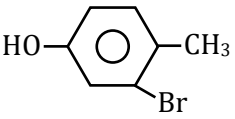
AIM : FREE EDUCATION TO ALL APUL

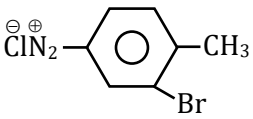
14. 

What is product B ?

(1) 

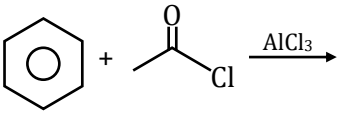
(2) 

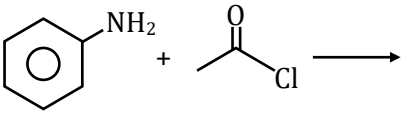
(3) 

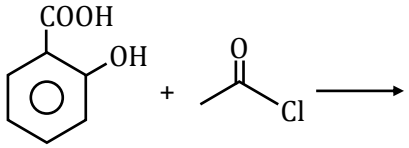
(4) 

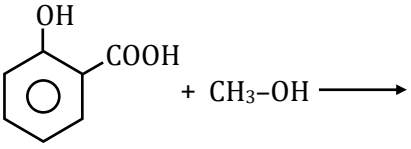
CAM190

15. In which reaction acetylation does not take place ?

(1) 

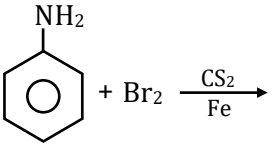
(2) 

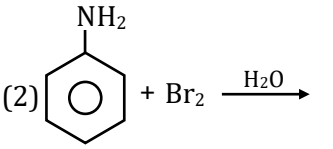
(3) 

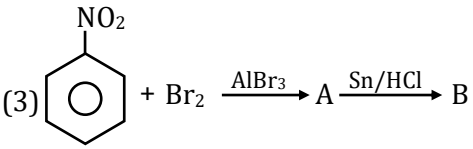
(4) 

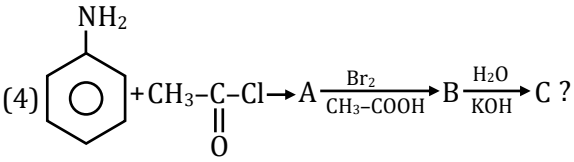
CAM191

16. Which is better method to obtained para bromo aniline in good yield ?

(1) 

(2) 

(3) 

(4) 

CAM192

17. Acetylation of Benzene is known as ?

(1) Gattermann reaction

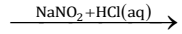
(2) Friedel Craft reaction

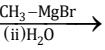
(3) Schotten Baumann reaction

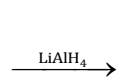
(4) Sandmeyer reaction

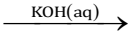
CAM193

18. Alcohol is not formed as product in

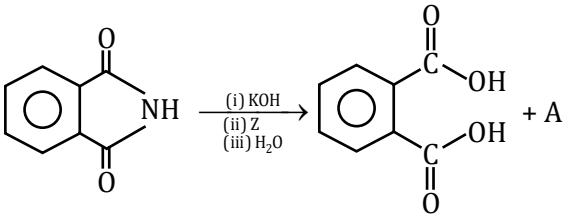
(1) 

(2) 

(3) 

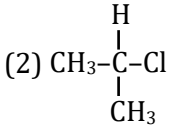
(4) 

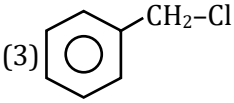
CAM194

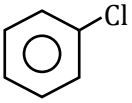
19. 

A is Primary amine
Z can not be :

(1) CH_3-Cl

(2) 

(3) 

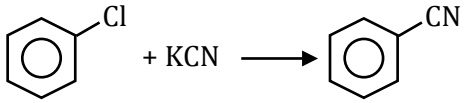
(4) 

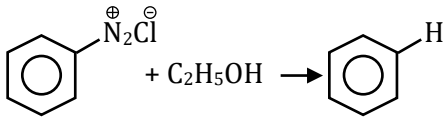
CAM195

AIM : FREE EDUCATION TO ALL APUL

20. Which reaction is possible ?

(1) $CH_3–CH_2–OH + NaBr \longrightarrow CH_3–CH_2–Br$

(2) 

(3) 

(4) All of them

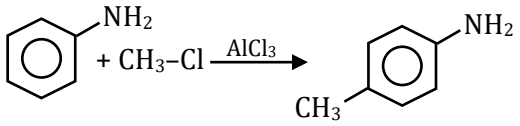
CAM196

21. Which reaction is not possible regarding major product ?

(1) $CH_3—CH_2—CH_2—OH + H–Cl \longrightarrow$

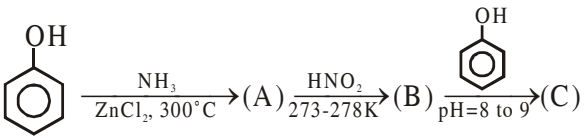
$CH_3—CH_2—CH_2—Cl$

(2) $CH_3–CH_2–\underset{\underset{OH}{|}}{CH}–CH_3 \xrightarrow[140°]{\text{Conc. } H_2SO_4} CH_3–CH=CH–CH_3$

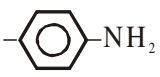
(3) 

(4) All of these

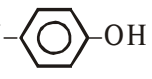
CAM197

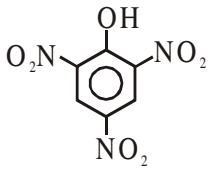
22. 

Final product ?

(1) 

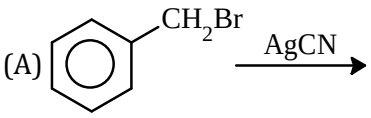
(2) Ph–N=N–Ph

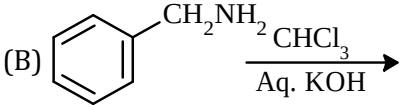
(3) 

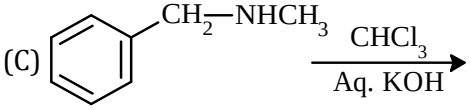
(4) 

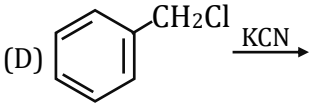
CAM198

23. Benzyl isocyanide can be obtained by:

(A) 

(B) 

(C) 

(D) 

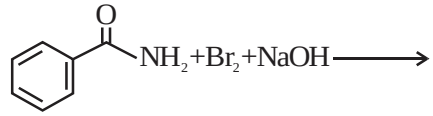
Choose the correct answer from the options given below :

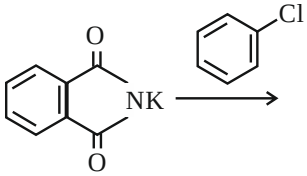
(1) A and D (2) Only B

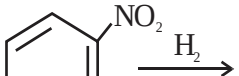
(3) A and B (4) B and C

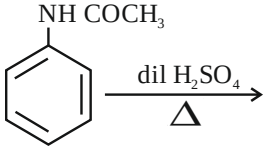
CAM199

24. How many of the transformation given below would result in aromatic amines?









(1) 1 (2) 2 (3) 3 (4) 4

CAM200

25. Which of the following conversion is incorrect:

(1) $CH_3CN \xrightarrow[\text{or LAH}]{Na/C_2H_5OH} CH_3CH_2NH_2$

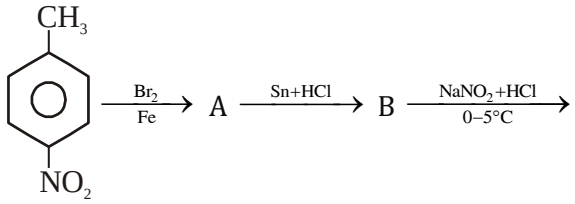
(2) $CH_3CONH_2 \xrightarrow{Br_2/KOH} CH_3CH_2NH_2$

(3) $CH_3CONH_2 \xrightarrow{H_3O^{\oplus}} CH_3COOH$

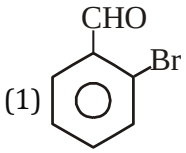
(4) $CH_3CONH_2 \xrightarrow{P_2O_5} CH_3CN$

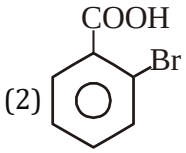
CAM201

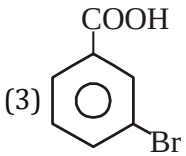
26.

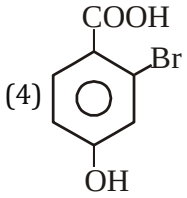


End product E will be :

(1) 

(2) 

(3) 

(4) 

CAM202

Exercise-II — ANSWER KEY

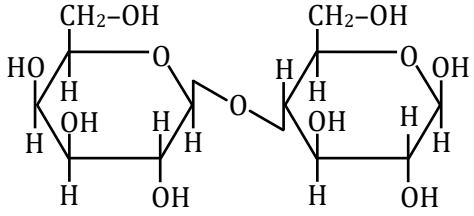
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**3**	**2**	**2**	**1**	**2**	**2**	**4**	**2**	**3**	**1**	**3**	**2**	**2**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26				
Answer	**4**	**2**	**3**	**4**	**3**	**4**	**3**	**3**	**3**	**2**	**2**				

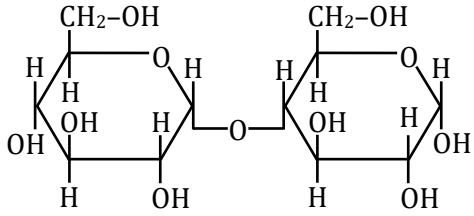
AIM : FREE EDUCATION TO ALL APUL

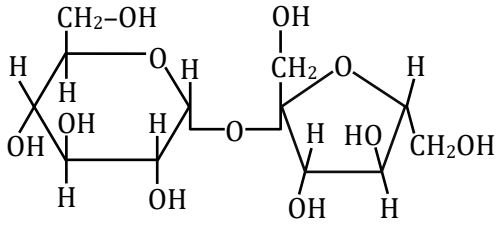
AIM : FREE EDUCATION TO ALL APUL

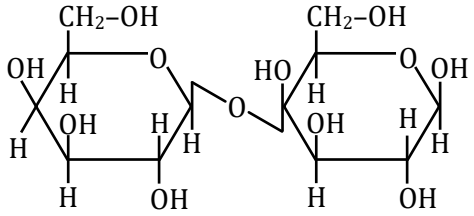
Exercise - I Biomolecules

1. Which of the following disaccharide is a non-reducing sugar :

(1) 

(2) 

(3) 

(4) 

CBM128

2. Lysine, $H_2N-(CH_2)_4-\underset{\underset{NH_2}{|}}{CH}-COOH$ is :

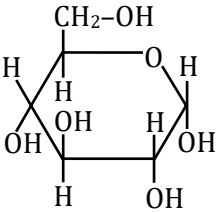
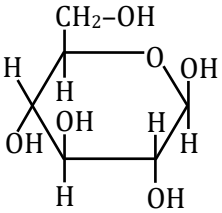
(a) α-amino acid
(b) Basic amino acid
(c) Amino acid synthesis in body
(d) β-Amino acid
(1) a, b, c (2) b, c (3) b, d (4) a, b

CBM129

3. Which of the following reaction of glucose can be explained only by its cyclic structure :
(1) Glucose forms pentaacetate
(2) Glucose reacts with hydroxylamine to form an oxime
(3) Pentacetate of glucose does not react with hydroxylamine
(4) Glucose is oxidised by Bromine water to gluconic acid

CBM130

4. Relation between (A) and (B) is :

(A)  (B) 

(1) Enantiomers (2) Anomers
(3) Positional isomers (4) Functional isomers

CBM131

5. Which of the following show mutarotation:
(a) Glucose (b) Sucrose
(c) Maltose (d) Galactose
(1) a, b, c (2) b, c (3) a, c, d (4) b, d

CBM132

6. Glucose and mannose are :
(1) Anomers (2) Position isomers
(3) Functional isomers (4) Epimers

CBM133

7. Invert sugar is :
(1) Sucrose (2) Fructose
(3) Glucose (4) Galactose

CBM134

8. In purine nucleoside, C-1 of sugar form glycosidic linkage with which position of purine:
(1) 1 (2) 2 (3) 9 (4) 8

CBM135

9. Which of the following is sulphur containing amino acid :
(a) Methionine (b) Cysteine
(c) Leucine (d) Glutamic acid
(1) a, b, c (2) a, b (3) b, c (4) c, d

CBM136

10. A metal present in vitamin B-12 is:
(1) Aluminium (2) Zinc
(3) Iron (4) Cobalt

CBM137

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**3**	**4**	**3**	**2**	**3**	**4**	**1**	**3**	**2**	**4**

Exercise - II — Biomolecules

1. Which of the following hormones contains iodine
(1) Thyroxine (2) Insulin
(3) Testosterone (4) Adrenaline

CBM138

2. α-D-(+)glucose and β-D-(+)glucose are
(1) position isomers (2) anomers
(3) enantiomers (4) conformers

CBM139

3. Correct statement for α-D-glucose is
(1) Can show mutarotation
(2) Does not give 2,4-DNP test
(3) Shows reducing property with Tollens' reagent
(4) All of these

CBM140

AIM : FREE EDUCATION TO ALL APUL

Exercise-II — ANSWER KEY

Question	1	2	3
Answer	**1**	**2**	**4**